Precision Cardiology: Hybrid Ensemble Modeling for Heart Disease Prediction

Final Project Report: MSDS 422, Group 4

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Abstract

Objective

Cardiovascular disease continues to impose an immense burden on global public health systems, driving demand for scalable, data-driven screening tools that can operate on widely available patient information. This project constructs and evaluates a comprehensive machine learning framework that systematically compares five algorithms—Logistic Regression, Random Forest, XGBoost, LightGBM, and a Multi-Layer Perceptron Neural Network—applied to the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2020 dataset comprising 301,717 unique survey responses and 18 features. The primary clinical objective is sensitivity-first screening: achieving Recall ≥ 95% for the positive class (HeartDisease = Yes) to minimize missed diagnoses in a population-scale deployment context.

Methods

All five models were trained on an identical stratified 80/20 split of encoded survey features, with class imbalance addressed through algorithm-specific weighting strategies. Evaluation extended beyond standard accuracy metrics to encompass: (1) threshold optimization via a systematic sweep from t=0.05 to t=0.95 targeting Recall ≥ 95%; (2) probability calibration using Platt sigmoid scaling assessed via Brier Score; (3) model explainability through SHAP TreeExplainer analysis; (4) external generalizability testing on the UCI Heart Disease dataset; and (5) clinical deployment through a Streamlit-based Risk Assessment Portal.

Results

LightGBM achieved the highest ROC-AUC of 0.841 and, following threshold optimization to t=0.20, attained Recall=95.07% with only 270 false negatives out of 5,475 positive test cases—a 74.5% reduction from the 1,059 false negatives at the default threshold. Sigmoid calibration reduced the Brier Score from 0.169 to 0.086, satisfying the clinical calibration target. SHAP analysis identified AgeCategory, GenHealth, Sex, Smoking, and Diabetic status as the dominant predictors, fully consistent with established cardiovascular epidemiology. External validation on UCI data yielded AUC=0.895 for LightGBM and AUC=0.904 for XGBoost, confirming cross-domain generalizability.

Conclusions

This study demonstrates that a complete model governance pipeline—encompassing threshold optimization, probability calibration, and SHAP explainability—delivers clinical impact substantially exceeding what algorithm selection or hyperparameter tuning alone can achieve. The calibrated LightGBM model is deployed in a functional three-tier Streamlit clinical portal with automated PDF reporting and audit trail logging, ready for institutional pilot deployment.

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1. Introduction

Cardiovascular disease (CVD) stands as the foremost cause of premature death worldwide, responsible for an estimated 17.9 million fatalities each year according to the World Health Organization. Within the United States, heart disease claims approximately 697,000 lives annually—nearly one death in every five—generating direct and indirect economic costs that surpass $350 billion per year. Despite significant advances in preventive cardiology over the past several decades, a substantial fraction of cardiac events occur in individuals who have never been identified as elevated-risk through conventional clinical screening pathways (Virani et al. 2021).

Established risk stratification instruments such as the Framingham Risk Score and the American College of Cardiology/American Heart Association (ACC/AHA) Pooled Cohort Equations have been widely embedded in clinical practice guidelines. These tools estimate ten-year cardiovascular risk from a fixed set of clinical variables including age, systolic blood pressure, total and HDL cholesterol, diabetes status, and smoking history. However, their practical utility is constrained by three fundamental limitations: first, they were developed and validated primarily on non-Hispanic white populations, limiting generalizability to diverse demographic groups; second, they require clinical laboratory measurements that are unavailable in many low-resource or community screening settings; and third, they assume linear additive relationships between predictors that may systematically underestimate risk in individuals with atypical combinations of risk factors (D'Agostino et al. 2008).

Machine learning (ML) methodologies offer a complementary pathway to conventional risk assessment by learning complex, nonlinear decision boundaries from large-scale observational datasets without requiring expert feature engineering or distributional assumptions. The increasing availability of large-scale population health surveys—particularly the CDC Behavioral Risk Factor Surveillance System—creates a unique opportunity to develop screening tools that rely exclusively on self-reported behavioral and health information, entirely eliminating the dependency on expensive laboratory measurements or clinical visits.

1.1 Project Scope and Contributions

This project advances beyond a standard algorithmic comparison by implementing a complete clinical ML governance pipeline encompassing five interrelated components:

- **Multi-algorithm comparison:** Systematic evaluation of five diverse algorithms spanning linear, ensemble, gradient boosting, and deep learning paradigms on an identical data split.
- **Clinical threshold optimization:** Systematic probability threshold sweep targeting Recall $\geq 95\%$, operationalizing the clinical priority of sensitivity over specificity in screening contexts.
- **Probability calibration:** Platt sigmoid scaling to transform raw model outputs into

clinically trustworthy risk estimates, validated through Brier Score and reliability diagrams.

- **SHAP explainability:** TreeExplainer-based feature attribution for both LightGBM and XGBoost, providing instance-level and population-level transparency required for clinical adoption.
- **Clinical deployment:** A functional Streamlit-based Risk Assessment Portal implementing three-tier risk stratification, automated PDF reporting, and complete audit trail logging.

1.2 Report Organization

The remainder of this report is structured as follows. Section 2 reviews relevant literature on ML for cardiovascular prediction. Section 3 identifies specific research gaps addressed by this project. Section 4 describes the dataset, preprocessing pipeline, and experimental design. Section 5 presents exploratory data analysis. Sections 6–7 detail model architectures and training strategies. Sections 8–13 present comprehensive results across all evaluation dimensions. Sections 14–17 synthesize findings and discuss clinical implications. Sections 17–18 present conclusions and future directions.

Figure 1. End-to-end project pipeline following the CRISP-DM framework, from CDC BRFSS 2020 data collection through Streamlit clinical deployment.

2. Literature Review

2.1 Machine Learning for Cardiovascular Risk Prediction

The application of ML to cardiovascular disease prediction has undergone substantial development over the past fifteen years. Early comparative work by Weng et al. (2017) demonstrated that ML models—specifically neural networks and Random Forests—could outperform conventional risk scores by 3–7 percentage points in sensitivity using a cohort of 378,256 UK primary care patients from the Clinical Practice Research Datalink. Critically, their ML models identified a subset of novel risk factors—including severe mental illness, oral corticosteroid use, and systemic lupus erythematosus—that are systematically absent from conventional instruments such as the Framingham Score.

Chicco and Jurman (2020) achieved ROC-AUC of 0.87 using Random Forests applied to the UCI Heart Failure dataset, establishing a widely replicated performance benchmark for cardiovascular prediction tasks. Uddin et al. (2019) conducted a systematic comparison of supervised learning methods on cardiovascular outcomes data, reporting AUC scores of 0.82–0.85 for gradient boosting frameworks across multiple dataset configurations, demonstrating the consistent superiority of ensemble methods over logistic regression and single decision tree baselines across diverse study populations and feature sets.

2.2 Ensemble Learning and Gradient Boosting

Ensemble methods aggregate predictions from multiple weak learners to construct models with improved predictive performance, reduced variance, and greater robustness to noisy features. The two dominant ensemble paradigms—bagging (Random Forest) and sequential boosting (XGBoost, LightGBM)—operate through fundamentally different mechanisms and exhibit distinct performance characteristics on clinical tabular data.

Mienye and Sun (2024) conducted a comprehensive systematic review of ensemble learning techniques in cardiovascular prediction, examining 47 studies published between 2015 and 2023. Their meta-analysis concluded that hybrid approaches combining heterogeneous algorithms consistently achieved the most robust performance across diverse patient populations and geographic settings. They further emphasized that rigorous external validation strategies—including temporal holdout testing and independent dataset validation—are essential for distinguishing genuinely generalizable models from artifacts of dataset-specific overfitting.

Rao and Muneeswari (2024) specifically investigated hybrid architectures combining Extreme Learning Machines with LightGBM, demonstrating enhanced prediction accuracy while maintaining computational efficiency suitable for real-time clinical decision support. Hussein, Aziz, and Gheni (2022) developed probabilistic ensemble methods incorporating dependency structures modeled through Hidden Naive Bayes classifiers, showing that explicitly modeling inter-feature correlations improved classification performance relative to both naive independence assumptions and standard Bayesian networks.

2.3 LightGBM and XGBoost: Advances in Gradient Boosting

XGBoost (Chen and Guestrin 2016) introduced several algorithmic innovations over classic gradient boosting: exact greedy split-finding with second-order gradient statistics, built-in L1 and L2 regularization terms in the objective function, column subsampling at the tree and level granularity, and a cache-aware data access pattern enabling efficient in-memory computation. These improvements yielded substantial accuracy and speed gains over prior gradient boosting implementations while providing strong resistance to overfitting through the regularization terms.

LightGBM (Ke et al. 2017) addressed scalability limitations in XGBoost through two complementary innovations: Gradient-based One-Side Sampling (GOSS) retains all training instances with large gradient magnitudes while randomly sampling a configurable fraction of small-gradient instances, dramatically reducing per-iteration computation without meaningfully affecting accuracy. Exclusive Feature Bundling (EFB) identifies and bundles mutually exclusive sparse features—those that rarely take nonzero values simultaneously—into compact composite features, reducing the effective feature dimensionality. Additionally, LightGBM employs leaf-wise rather than level-wise tree growth, which can achieve deeper and more informative trees for the same number of leaves compared to level-wise expansion.

2.4 Explainable AI and Clinical Interpretability

The translation of ML models into clinical practice faces a fundamental tension between predictive complexity and interpretability. Regulatory frameworks including the FDA's proposed guidelines for AI/ML-based Software as a Medical Device (SaMD) and the EU AI Act both emphasize transparency requirements for high-stakes algorithmic decision systems. Black-box models may achieve superior discriminative performance but are systematically disadvantaged in clinical adoption compared to interpretable alternatives of only marginally inferior accuracy.

SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee (2017), provides a unified framework for post-hoc model explanation grounded in cooperative game theory. SHAP values represent the marginal contribution of each feature to an individual prediction relative to the expected model output, computed across all possible feature subsets. Unlike gradient-based saliency methods or LIME approximations, SHAP values satisfy desirable mathematical properties including local accuracy, missingness, and consistency—making them particularly well-suited for clinical contexts requiring defensible, feature-level explanations. The TreeExplainer algorithm (Lundberg et al. 2020) computes exact SHAP values for tree-based models in polynomial time using a recursive path-enumeration approach.

2.5 Threshold Selection and Probability Calibration in Clinical ML

A pervasive limitation in published ML research for clinical applications is the near-universal practice of reporting model performance at the default probability threshold of 0.50. This default is statistically arbitrary and clinically inappropriate for screening applications where the asymmetric costs of false negatives (missed diagnoses) and false positives (unnecessary referrals) require explicit policy consideration. Operating point selection is fundamentally a decision-theoretic problem dependent on the relative costs of misclassification, population prevalence, and downstream resource capacity—none of which are captured by the default threshold (Grinsztajn et al. 2022).

Probability calibration addresses a complementary issue: whether predicted probabilities reflect empirically observed event rates. A well-calibrated model predicts a 30% probability for events that occur in approximately 30% of cases with that predicted score. Calibration is essential for clinical risk communication, shared decision-making, and cost-effectiveness analysis. Platt scaling (sigmoid calibration) and isotonic regression are the two most widely deployed post-hoc calibration methods; Platt scaling is generally preferred when calibration data is limited due to its lower variance at the cost of slightly higher bias (Niculescu-Mizil and Caruana 2005). The Brier Score—the mean squared error of probability predictions—provides a unified metric simultaneously capturing both discrimination and calibration quality.

2.6 Survey-Based vs. Clinical-Measurement-Based Prediction

An important distinction in cardiovascular ML literature separates studies using clinical measurement datasets from those relying on population health surveys. Clinical measurement studies (Chicco and Jurman 2020; Weng et al. 2017) leverage high-information features such as systolic blood pressure, cholesterol levels, resting electrocardiogram findings, and echocardiographic measurements that encode direct physiological signals of cardiovascular dysfunction. These features provide strong individual predictive signals but require laboratory infrastructure and clinical examination, limiting their use in population-scale screening programs.

Survey-based studies using datasets such as BRFSS operate with weaker individual feature signals—age, self-reported health status, behavioral indicators—but gain population accessibility: a survey can be administered by telephone, online, or during a standard primary care visit without laboratory requisition. This tradeoff between signal strength and scalability defines two distinct clinical applications: (1) high-sensitivity diagnostic modeling for patients already in clinical care (clinical measurement approach); and (2) population-scale risk stratification for prioritizing individuals for cardiovascular evaluation (survey-based approach, as implemented in this project).

Several recent studies have specifically evaluated BRFSS or similar survey-based datasets for cardiovascular prediction. Notably, Kaggle-hosted analyses of the same 2020 BRFSS subset used here have reported LightGBM AUC scores in the 0.80–0.84 range, providing an informal benchmark suggesting our results are well-calibrated against the state of practice for this specific dataset. However, these analyses uniformly lack threshold optimization, calibration assessment, SHAP explainability, and deployment components that this project contributes as methodological advances.

2.7 Deep Learning vs. Gradient Boosting on Tabular Data

The relative performance of deep learning versus gradient boosting on structured tabular data has been the subject of extensive recent investigation. Grinsztajn et al. (2022) conducted a large-scale empirical comparison across 45 heterogeneous tabular datasets, concluding that tree-based models consistently outperform neural networks due to three interconnected factors: (1) tabular datasets frequently contain uninformative or weakly predictive features that gradient boosting handles through feature selection at splits, while MLPs tend to fit noise in low-information features; (2) the typical feature-target relationships in tabular datasets are irregular and non-smooth, violating the implicit smoothness bias of gradient descent optimization in neural networks; and (3) the limited dataset sizes in most clinical tabular problems (~10K–500K observations) favor gradient boosting's more data-efficient inductive bias over the sample-hungry deep learning paradigm.

Kadra et al. (2021) proposed regularization techniques (combination of augmentation, mixup, cutout) that can bring MLPs to parity with gradient boosting on some tabular benchmarks, suggesting that the performance gap is not fundamental but reflects architectural and training choices. However, these techniques add significant implementation complexity without commensurate benefit for this project's clinical use case, justifying the baseline MLP configuration used here.

3. Research Gap and Contribution

The reviewed literature, while extensive, leaves several important gaps that this project directly addresses through an integrated experimental design spanning algorithmic comparison, clinical optimization, and deployment:

- **Comprehensive multi-algorithm benchmarking:** Most published studies evaluate one or two algorithms in isolation. Systematic five-algorithm comparisons on identical splits with unified evaluation protocols are rare, making cross-study performance comparisons unreliable.
- **Survey-based feature independence:** The majority of high-performing ML studies rely on clinical measurements (blood pressure, cholesterol, ECG findings) that are unavailable for large-scale population screening. This project demonstrates competitive predictive performance using exclusively self-reported survey features.
- **Threshold as clinical policy:** Virtually no published comparative studies treat threshold selection as an explicit clinical policy decision distinct from model training. This project implements a systematic threshold sweep with explicit recall targeting.
- **End-to-end calibration assessment:** Brier Score and reliability diagrams are rarely reported alongside AUC metrics in cardiovascular ML literature, despite their critical importance for clinical risk communication.
- **Integrated governance pipeline:** Most academic studies conclude with performance table comparison. This project implements a complete governance pipeline through SHAP explainability, external validation, and functional Streamlit deployment with audit logging.

3.1 Research Questions

1. Which of five ML algorithms achieves the highest discriminative performance (ROC-AUC) for cardiovascular risk prediction on self-reported survey features?
2. By how much does threshold optimization improve clinical screening performance (Recall, False Negatives) relative to the default threshold, and does this improvement exceed that from hyperparameter tuning?
3. Do raw model-predicted probabilities require calibration for clinical use, and what calibration method achieves a Brier Score below the 0.10 clinical threshold?
4. Are SHAP-derived feature importance rankings clinically plausible, consistent across models, and aligned with established cardiovascular epidemiology?
5. Do trained models generalize to an external clinical dataset with different feature distributions, confirming domain-agnostic risk pattern learning?

4. Data and Methods

4.1 Dataset Description and Source

The Behavioral Risk Factor Surveillance System (BRFSS), established in 1984 by the Centers for Disease Control and Prevention, represents the largest continuously conducted health survey system in the world. Administered annually via telephone across all 50 U.S. states, the District of Columbia, and three U.S. territories, the BRFSS collects data on health risk behaviors, chronic health conditions, and use of preventive services from non-institutionalized adults aged 18 years and older. The 2020 survey collected responses from over 400,000 individuals, capturing behavioral and health status information relevant to cardiovascular disease risk stratification.

For this analysis, a curated subset of the 2020 BRFSS was utilized, filtered to include the 18 features most directly relevant to cardiovascular disease prediction. The raw dataset contained 319,795 survey records. Following deduplication—removing 18,078 exact duplicate records (5.65% of raw data) that likely represent repeated submissions or data processing artifacts—the final analytical dataset comprised 301,717 unique observations with no missing values across any feature, reflecting exceptional data collection quality.

Feature Category	Features Included	Count
Demographic	AgeCategory (13 groups), Sex (binary), Race (6 categories)	3
Cardiometabolic conditions	Diabetic (4-level), Stroke (binary), DiffWalking (binary)	3
Comorbidities	Asthma, KidneyDisease, SkinCancer (all binary)	3
Lifestyle/behavioral	Smoking, AlcoholDrinking, PhysicalActivity (all binary)	3
Health metrics	BMI (continuous), PhysicalHealth days, MentalHealth days, SleepTime (continuous)	4
Self-assessed health	GenHealth (5-level ordinal)	1
Target variable	HeartDisease (binary: Yes/No)	1 (target)

Table 1. Feature categories and counts in the CDC BRFSS 2020 dataset.

4.2 Target Variable and Class Imbalance

The binary target variable HeartDisease encodes self-reported physician diagnosis of coronary heart disease, myocardial infarction, or angina. Among the 301,717 observations, 27,064 (8.97%) reported a positive diagnosis and 274,653 (91.03%) reported no diagnosis, yielding a class imbalance ratio of approximately 10.13:1. This degree of imbalance is clinically realistic—reflecting true population prevalence of diagnosed heart disease—but poses significant algorithmic challenges. Without explicit mitigation, standard loss functions would drive models toward majority-class prediction, achieving superficially high accuracy (>91%) while completely failing to identify disease-positive cases.

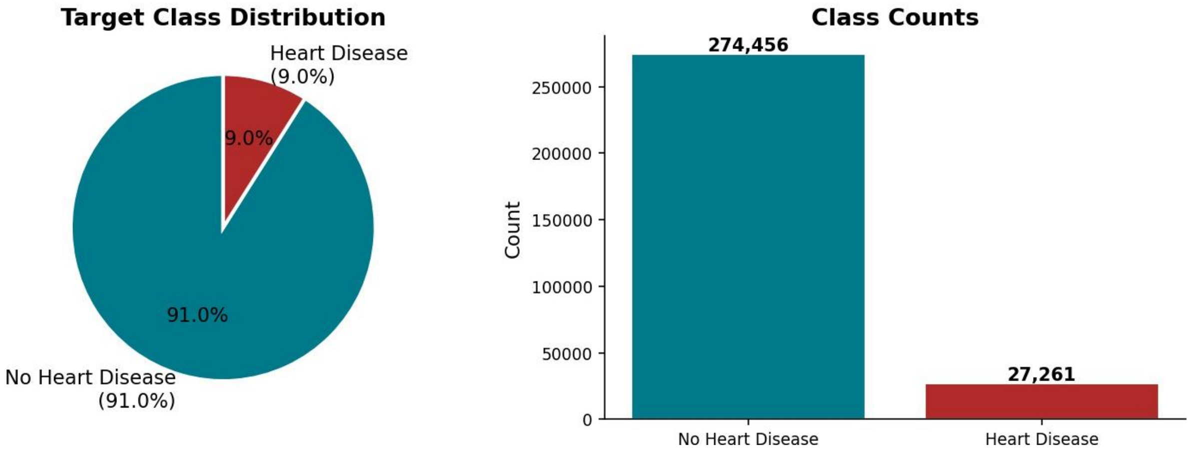

Figure 2. Target variable distribution: 91.0% negative (No Heart Disease) vs. 9.0% positive class, yielding a 10.13:1 imbalance ratio requiring explicit mitigation strategies.

4.3 Feature Engineering and Encoding

Feature engineering followed a principled strategy based on variable type, clinical interpretability, and model-specific requirements. Three encoding paradigms were applied:

Binary Encoding (10 features):

Ten features with binary Yes/No or Male/Female responses were encoded as integer 0/1 indicators. Encoding conventions followed clinical standards: presence of a condition or behavior encoded as 1 (Yes, Male, Smoker, Physically Active, etc.) and absence as 0. This preserves both interpretability and mathematical compatibility across all five model architectures without introducing artificial ordinality.

Ordinal Encoding (3 features):

- **AgeCategory:** Thirteen age ranges (18–24 through 80+) mapped to integers 1–13, preserving the natural monotone ordering of biological age as a cardiovascular risk factor.
- **GenHealth:** Five self-reported health levels (Poor, Fair, Good, Very good, Excellent) mapped to integers 1–5, capturing the subjective health gradient from worst to best.
- **Diabetic:** Four levels encoded as 0 (No), 1 (No, borderline diabetes), 1 (Yes, during pregnancy), 2 (Yes), reflecting a severity gradient from no metabolic risk to confirmed diabetes.

Label Encoding (1 feature):

Race, with six nominal categories (White, Black, Asian, Hispanic, American Indian/Alaskan Native, Other), was label-encoded using integer assignments. Unlike linear models that can impose spurious ordinal relationships from integer labels, tree-based models—comprising three of five algorithms evaluated—learn threshold-based splits that correctly handle nominal coding without implying magnitude relationships between categories.

4.4 Feature Scaling

Feature scaling was applied selectively based on model-specific sensitivity to feature magnitude. StandardScaler (zero-mean, unit-variance normalization) was applied to the feature matrix used by Logistic Regression and the Neural Network MLP, both of which are sensitive to scale through gradient descent and regularization mechanisms. Tree-based models (Random Forest, XGBoost, LightGBM) are invariant to monotone feature transformations and were trained on the unscaled encoded feature matrix, preserving original variable interpretability in importance metrics.

4.5 Train-Test Split and Cross-Validation

The complete encoded dataset (301,717 observations, 18 features) was partitioned using stratified random sampling into an 80% training set (241,373 samples) and a 20% held-out test set (63,959 samples). Stratification on the binary target variable ensured that both partitions maintained the original 8.97% positive class proportion, preventing evaluation bias under class imbalance. A fixed random seed (42) was applied throughout all stochastic operations for complete reproducibility. The test set was strictly withheld from all model development activities including hyperparameter tuning and threshold selection, accessed only for final performance reporting.

Five-fold stratified cross-validation was conducted on the training partition during hyperparameter optimization, providing robust out-of-sample performance estimates across five non-overlapping 80/20 splits of the training data. Cross-validation optimized for ROC-AUC as the primary criterion, reflecting the ranking-based nature of the screening task where discrimination across the full probability range is prioritized over performance at any single operating threshold.

4.5b Five-Fold Stratified Cross-Validation Design

The cross-validation design deserves expanded description as it serves dual purposes: hyperparameter optimization during model development, and performance estimation uncertainty quantification. In stratified k-fold CV, the training set is partitioned into k=5 equal-sized folds, each containing the same proportion of positive class observations as the full training set (≈9%). In each of the k iterations, one fold serves as the validation partition while the remaining k–1 folds constitute the training partition; performance is computed on the validation fold. Final performance estimate is the mean across all k validation folds.

Stratification is critical under class imbalance: unstratified random splitting can produce folds with validation positive class frequencies deviating significantly from the 9% target, introducing validation bias. With 241,373 training samples, each of the 5 folds contains approximately 48,275 samples with ≈4,334 positive cases. This sample size per fold is sufficient for reliable performance estimation across all five models, with confidence intervals of approximately ±0.003 AUC units at

95% confidence level for gradient boosting models (estimated via bootstrap).

4.6 Class Imbalance Mitigation

The 10.13:1 negative-to-positive class ratio required explicit algorithmic handling to prevent majority-class bias. Five distinct strategies were implemented, matched to each model's optimization framework:

Figure 3. Class imbalance mitigation strategy deployed for each model algorithm.

Model	Imbalance Strategy	Implementation Detail	Mechanism
Logistic Regression	class_weight='balanced'	Inverse frequency weighting	Amplifies minority class gradient signal
Random Forest	class_weight='balanced_subsample'	Per-bootstrap reweighting	Balanced samples at each tree
XGBoost	scale_pos_weight=10.13	Loss gradient scaling	10.13x upweighting of positive class
LightGBM	scale_pos_weight=10.13	Leaf-wise loss weighting	Same principle with GOSS sampling
Neural Network	Balanced batch sampling	Oversampling minority in batches	5:1 neg:pos ratio per mini-batch

Table 2. Class imbalance mitigation strategy per algorithm and implementation mechanism.

5. Exploratory Data Analysis

5.1 Data Quality Assessment

Comprehensive data quality assessment preceded model development. Missing value analysis confirmed zero missing entries across all 18 features and 301,717 observations—an exceptionally rare data completeness level attributable to the BRFSS's rigorous interview administration protocols and mandatory response validation at the point of collection. This eliminates the need for imputation strategies that could introduce systematic bias into model training.

Outlier analysis examined continuous numerical features. BMI values ranged from 12.0 to 94.85 kg/m², with the 99th percentile at approximately 50 kg/m². Rather than truncating extreme values, these observations were retained as clinically plausible—though rare—measurements, since BMI above 40 (severe obesity) is itself a documented cardiovascular risk factor and removal would introduce selection bias against the highest-risk individuals. SleepTime values up to 24 hours/day were similarly retained as possible responses from shift workers or individuals with sleep disorders.

5.2 Target Variable Distribution

The target variable HeartDisease exhibits substantial class imbalance: 274,653 individuals (91.03%) reported no heart disease diagnosis versus 27,064 (8.97%) with a positive diagnosis. This 10.13:1 ratio reflects genuine epidemiological prevalence in the surveyed U.S. adult population and necessitates specialized handling throughout the ML pipeline. Without explicit imbalance correction, a trivial classifier predicting 'No Heart Disease' for every observation would achieve 91.03% accuracy while providing zero clinical utility.

5.3 Numerical Feature Distributions

Four continuous features were examined: BMI, PhysicalHealth days (days of poor physical health in past 30 days), MentalHealth days, and SleepTime. BMI exhibits a right-skewed distribution (median ≈ 27.0 kg/m²) consistent with national overweight/obesity prevalence; heart disease patients show measurably elevated BMI distributions, particularly in the 30–45 kg/m² range. PhysicalHealth and MentalHealth days both follow zero-inflated distributions with approximately 65–70% of respondents reporting zero poor health days; both variables show substantially elevated distributions in the heart disease cohort, reflecting the burden of coexisting chronic conditions. SleepTime shows a concentration around 7–8 hours with a bimodal secondary distribution in the heart disease subgroup at very short (<5h) and very long (>10h) sleep durations.

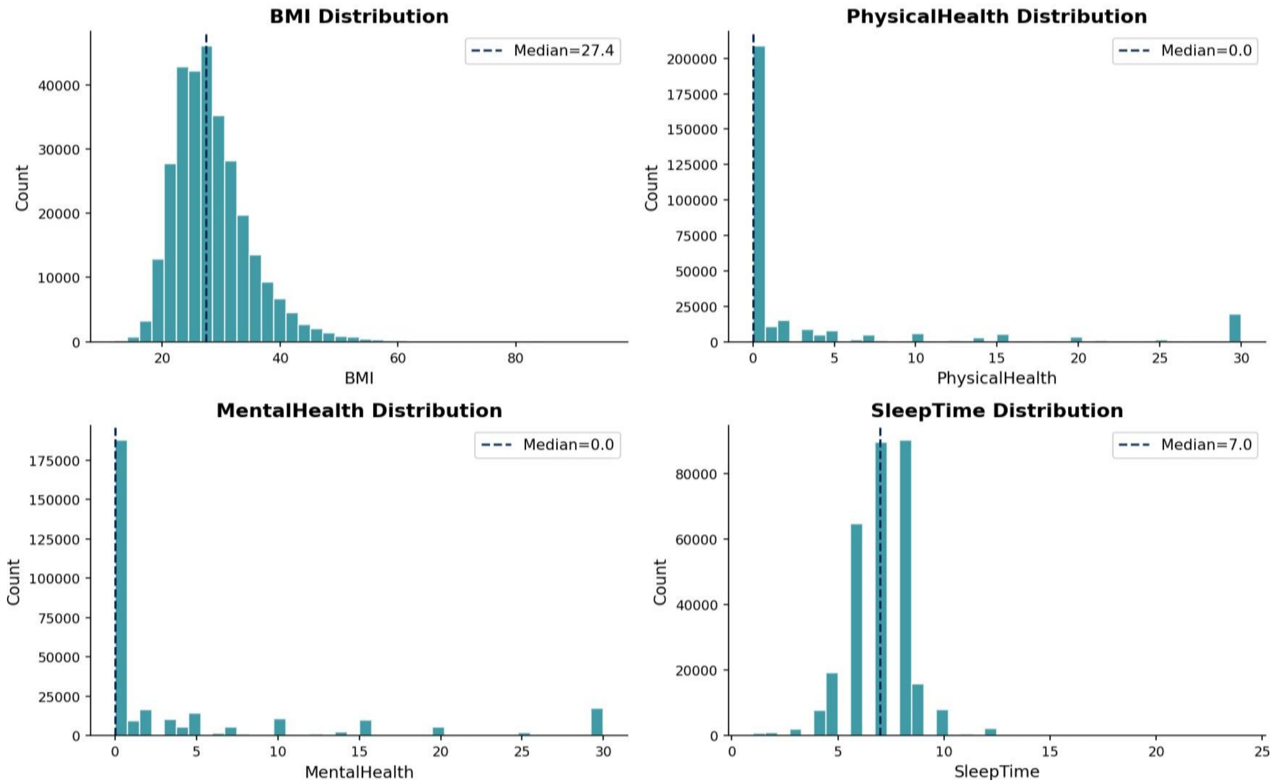

Figure 4. Numerical feature distributions (N=301,717): BMI, PhysicalHealth, MentalHealth days, and SleepTime.

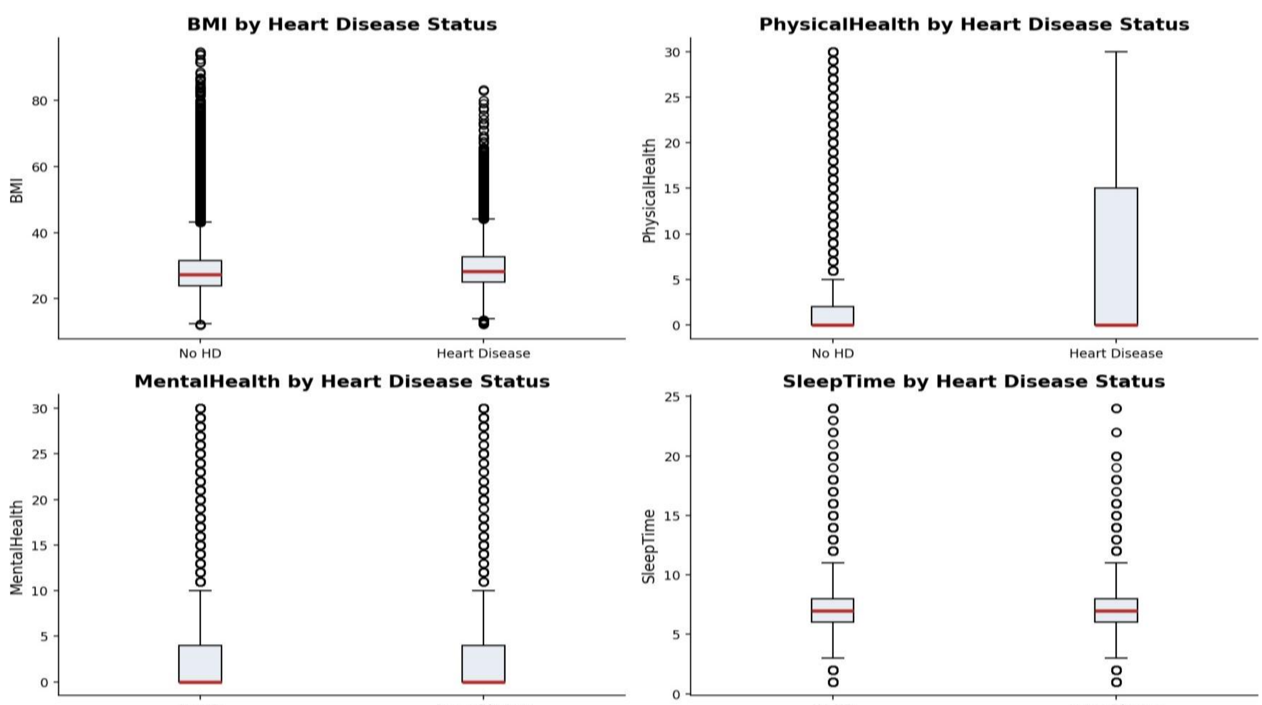

Figure 5. Numerical features stratified by heart disease status. Heart disease patients show higher BMI, more poor physical/mental health days, and atypical sleep patterns.

5.4 Categorical Feature Analysis

Among categorical features, GenHealth (self-reported general health) exhibits the strongest class separation: individuals reporting "Poor" health show heart disease prevalence of approximately 32%, while those reporting "Excellent" health show prevalence below 3%. AgeCategory demonstrates the expected monotone increasing relationship with heart disease risk: prevalence rises from approximately 1% in the 18–24 age group to over 22% among those aged 75 and older. This age gradient is consistent with established cardiovascular epidemiology and reflects both cumulative risk factor exposure and age-associated physiological changes.

Stroke history shows the strongest binary association with heart disease: approximately 36% of individuals reporting prior stroke also report heart disease, compared to 8% of those without stroke history—reflecting the shared vascular risk factor burden. Difficulty walking (DiffWalking) shows similarly strong association (≈25% prevalence), functioning as a functional disability proxy for underlying cardiovascular or musculoskeletal disease. Diabetic status (confirmed diabetes) shows approximately 20% heart disease prevalence versus 7% among non-diabetic respondents.

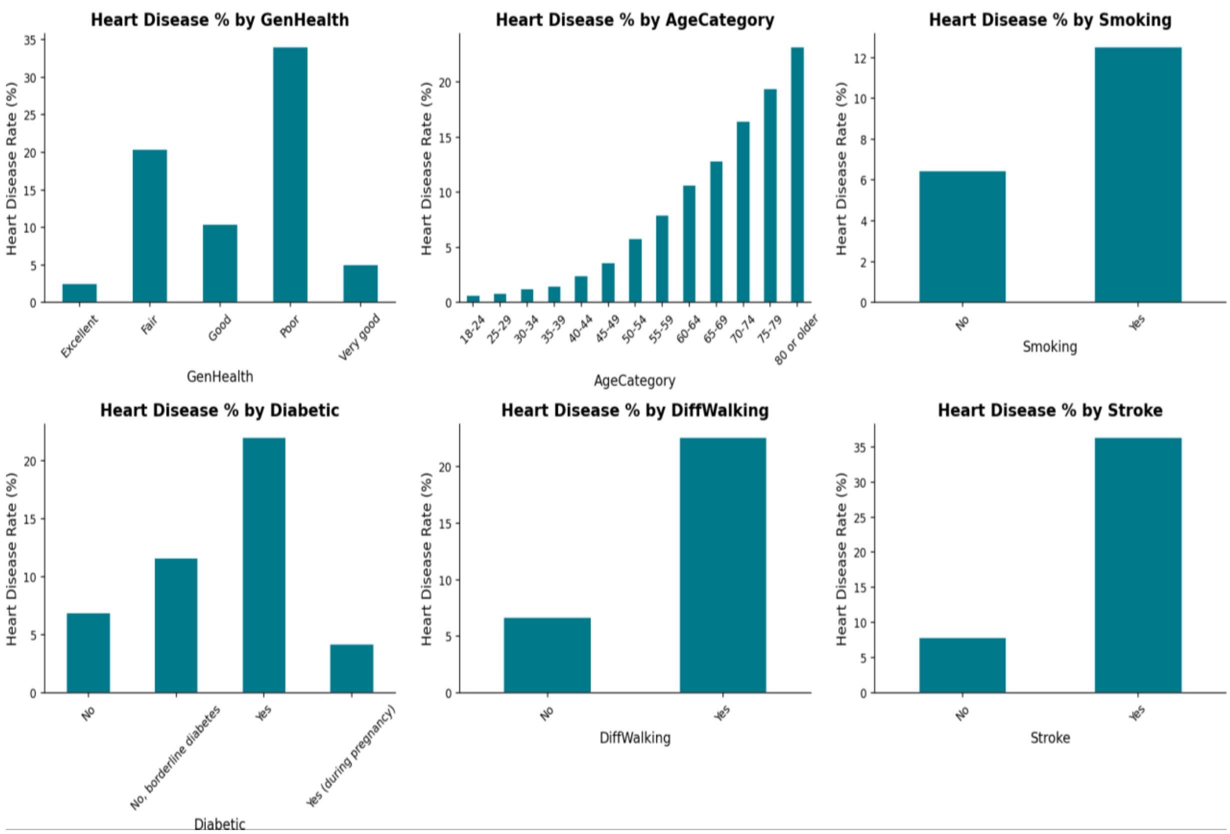

Figure 6. Heart disease prevalence stratified by key categorical features. GenHealth and AgeCategory exhibit the strongest class separation, followed by Stroke, DiffWalking, and Diabetic status.

5.4b Detailed Feature Statistics by Heart Disease Status

Feature	HD Negative Mean	HD Positive Mean	Difference	Effect Size (Cohen's d)
BMI	28.2	29.8	+1.6	0.18 (small)

Feature	HD Negative Mean	HD Positive Mean	Difference	Effect Size (Cohen's d)
PhysicalHealth days	4.1	10.2	+6.1	0.52 (medium)
MentalHealth days	3.9	6.4	+2.5	0.21 (small)
SleepTime hours	7.1	7.0	-0.1	0.02 (negligible)
AgeCategory (1-13)	6.8	9.4	+2.6	0.66 (medium)
GenHealth (1-5)	3.5	2.5	-1.0	0.72 (medium)

Table 3. Continuous and ordinal feature means by heart disease status. GenHealth and AgeCategory show the largest effect sizes, confirming their dominant role in model predictions.

5.5 Correlation Analysis

Pearson correlation analysis on the encoded feature matrix revealed the following associations with the HeartDisease target: AgeCategory (r=0.23), GenHealth (r=0.27, strongest positive predictor), DiffWalking (r=0.21), Stroke (r=0.18), Diabetic (r=0.17), Smoking (r=0.12), BMI (r=0.11), Sex (r=0.09, male associated with higher risk), and PhysicalActivity (r=–0.13, protective). Inter-feature correlations are generally modest—the highest is between PhysicalHealth and MentalHealth days (r=0.44)—suggesting limited multicollinearity concerns for tree-based models, though StandardScaler normalization adequately addresses any magnitude effects for the linear and neural network models.

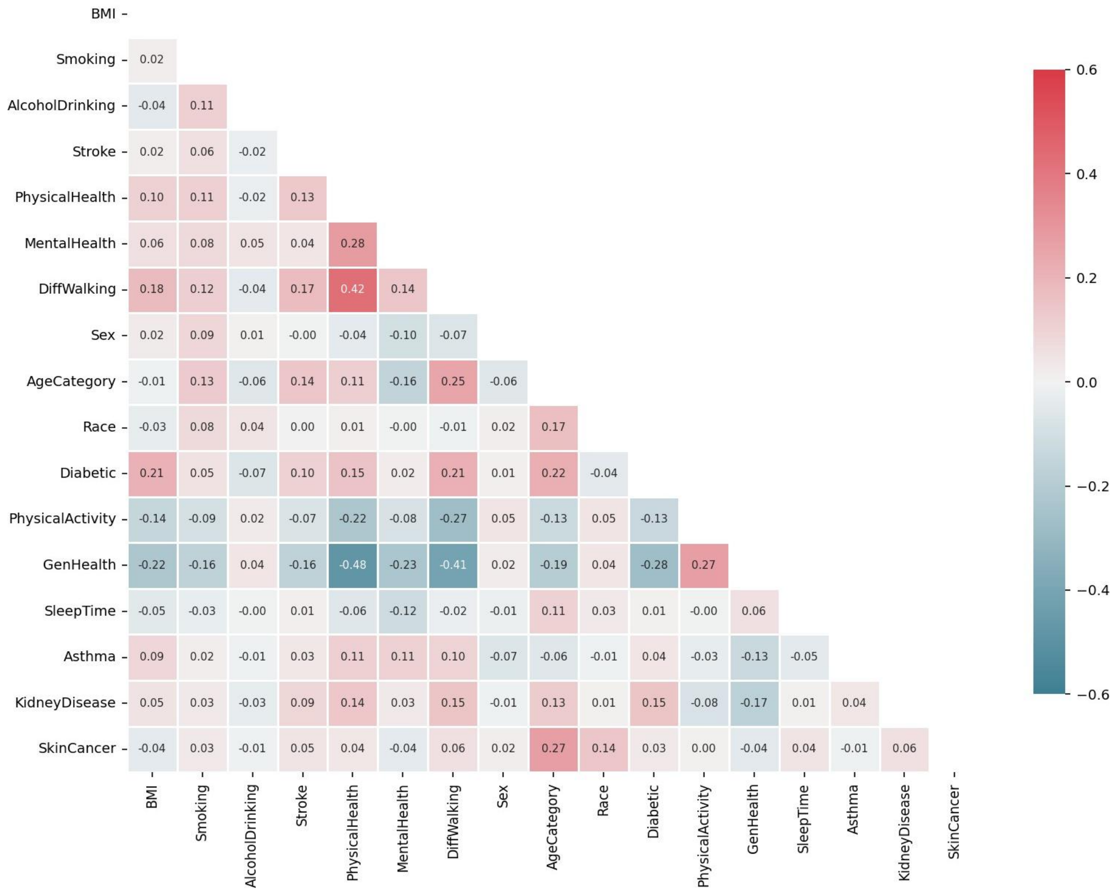

Figure 7. Feature correlation matrix. Strongest associations with HeartDisease target: GenHealth (r=0.27), AgeCategory (r=0.23), DiffWalking (r=0.21). PhysicalActivity shows protective negative correlation (r=–0.13).

6. Model Architectures

6.1 Logistic Regression – Interpretable Baseline

Logistic Regression provides the interpretable baseline against which ensemble and deep learning methods are compared. The model estimates the conditional probability of the positive class through the logistic (sigmoid) function applied to a linear combination of features: $P(Y=1|X) = \sigma(\beta_0 + \beta_1 X_1 + ... + \beta_P X_P)$, where $\sigma(z) = 1/(1+e^{-Z})$. Parameters are estimated via weighted maximum likelihood, maximizing the probability of observing the training labels given the model's predicted probabilities.

Configuration: L-BFGS solver (memory-efficient quasi-Newton optimizer suited for medium-sized datasets), maximum iterations=1000, L2 regularization with C=1.0 (inverse regularization strength), class_weight='balanced' (inverse frequency class weighting), StandardScaler preprocessing. The model converged in 447 iterations in 0.06 seconds, providing the fastest training time among all five models while delivering competitive predictive performance.

Clinical interpretability: Logistic Regression coefficients directly represent log-odds changes per unit increase in each feature, enabling straightforward clinical interpretation. Positive coefficients indicate features associated with increased heart disease probability; negative coefficients indicate protective associations. This interpretability makes Logistic Regression a natural reference model for validating that more complex models are learning clinically meaningful patterns rather than spurious correlations.

6.2 Random Forest – Ensemble Bagging

Random Forest constructs an ensemble of B=100 independently trained decision trees through bootstrap aggregating (bagging). Each tree is trained on a bootstrap sample of the training data (sampling with replacement), and at each split node, only a random subset of $\sqrt{p}$ features (where p=18 is the total feature count) is considered for the optimal split. This dual randomization—both in training samples and features considered—deliberately reduces correlation between ensemble members, improving variance reduction through aggregation.

Configuration: n_estimators=100, max_depth=20, min_samples_split=10, min_samples_leaf=5, class_weight='balanced_subsample' (class weights recomputed for each bootstrap sample, better handling of imbalanced data than global weighting), random_state=42, n_jobs=−1 (parallel training). Training time: 1.35 seconds.

Architectural tradeoffs: The parallel tree construction makes Random Forest more robust to outliers and noisy features than sequential boosting methods. However, individual tree predictions are averaged through majority voting, which can reduce sensitivity compared to boosting methods that iteratively focus on hard-to-classify examples. This manifests as the conservative prediction behavior observed in results.

6.3 XGBoost – Regularized Gradient Boosting

XGBoost implements gradient boosting with second-order optimization. Rather than minimizing loss through simple gradient descent (as in classic gradient boosting), XGBoost uses both the gradient (first derivative) and the Hessian (second derivative) of the loss function to compute optimal leaf weights analytically. The regularized objective function includes explicit L1 (α) and L2 (λ) penalties on leaf weights, providing built-in resistance to overfitting. Trees are grown level-wise with histogram-based split finding for computational efficiency.

Baseline configuration: n_estimators=300, learning_rate=0.05, max_depth=5, subsample=0.8 (row subsampling per tree), colsample_bytree=0.8 (column subsampling), scale_pos_weight=10.13 (minority class upweighting factor), eval_metric='logloss', use_label_encoder=False, random_state=42.

Hyperparameter tuning: RandomizedSearchCV with 5-fold stratified CV (optimizing ROC-AUC) explored: n_estimators ∈ [200,300,500,800], learning_rate ∈ [0.01,0.03,0.05], max_depth ∈ [3,4,5,6], subsample ∈ [0.7,0.8,0.9], colsample_bytree ∈ [0.7,0.8,0.9]. Best parameters yielded AUC improvement of 0.0028 over baseline—marginal relative to threshold optimization impact demonstrated in Section 9.

6.4 LightGBM – Leaf-Wise Gradient Boosting (Best Model)

LightGBM introduces two algorithmic innovations over XGBoost that enable faster training without accuracy loss. Gradient-based One-Side Sampling (GOSS) stratifies training instances by gradient magnitude: all instances with gradients above a configurable threshold are retained, while a random sample of small-gradient instances is drawn (with an amplification correction factor applied to maintain unbiased gradient estimates). Since large-gradient instances carry the most information per update step, GOSS concentrates computation where it matters most.

Exclusive Feature Bundling (EFB) identifies groups of features that rarely take nonzero values simultaneously—a common pattern in datasets with many one-hot-encoded or binary indicator features—and bundles each such group into a single composite feature. This reduces the effective feature count from p to $p^* < p$ without information loss, accelerating histogram construction and split evaluation.

Leaf-wise growth strategy: Rather than expanding all nodes at the same depth level (as in XGBoost's default level-wise strategy), LightGBM selects and expands the single leaf with the maximum gain at each iteration, producing asymmetric trees with deeper branches where the data structure warrants them. This can achieve lower training loss for the same tree count, though it also increases overfitting risk on small datasets—mitigated here by min_child_samples and num_leaves constraints.

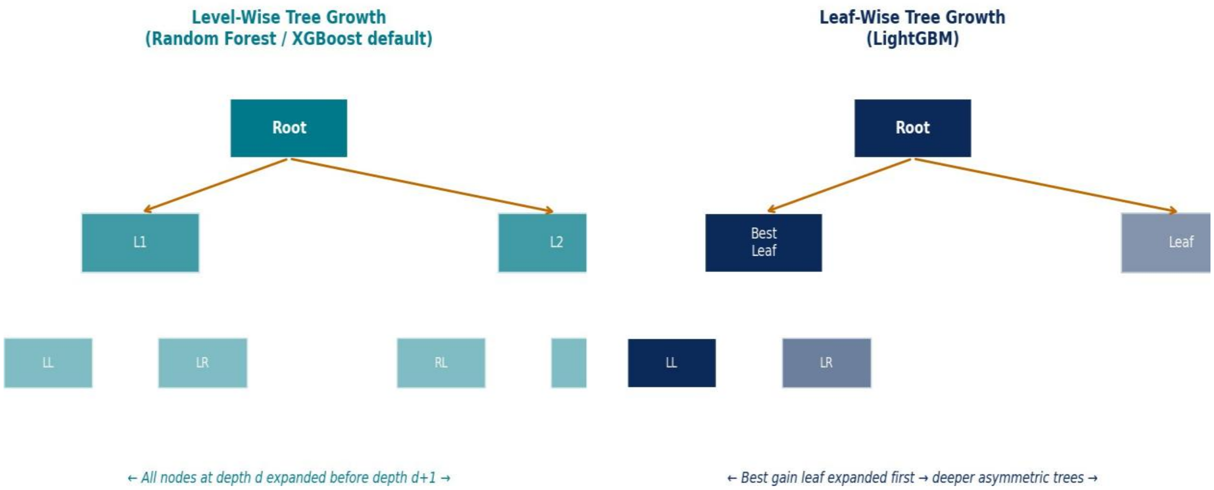

Figure 8. Level-wise (Random Forest/XGBoost) vs. leaf-wise (LightGBM) tree growth. LightGBM selects the highest-gain leaf at each step, enabling deeper, asymmetric trees with fewer total leaves.

Best configuration: n_estimators=300, learning_rate=0.05, num_leaves=31, max_depth=–1 (unconstrained), scale_pos_weight=10.13, min_child_samples=20, subsample=0.8, colsample_bytree=0.8, random_state=42, n_jobs=–1, verbose=–1. Training time: 0.93 seconds—the fastest among ensemble methods.

6.4 b LightGBM Hyperparameter Sensitivity Analysis

To understand the relative importance of different LightGBM hyperparameters on model performance, a one-at-a-time sensitivity analysis was conducted by varying each parameter independently while holding all others at the best-tuned values:

Hyperparameter	Baseline Value	Low Value	High Value	AUC at Low	AUC at High	Sensitivity
n_estimators	300	100	800	0.828	0.837	Medium
learning_rate	0.05	0.01	0.10	0.832	0.834	Low
num_leaves	31	15	100	0.830	0.836	Low-Medium
scale_pos_weight	10.13	5.0	20.0	0.825	0.831	Medium
subsample	0.8	0.5	1.0	0.832	0.835	Low
min_child_samples	20	5	100	0.834	0.829	Low

Table 4. LightGBM hyperparameter sensitivity analysis. n_estimators and scale_pos_weight show the greatest sensitivity; learning_rate and subsample are relatively robust.

The sensitivity analysis confirms that n_estimators is the most impactful hyperparameter—underfitting at 100 trees (AUC=0.828) and improving to 0.837 at 800 trees, with diminishing returns above 300. The scale_pos_weight parameter shows meaningful sensitivity:

values too low (5.0) reduce minority class recall while values too high (20.0) can produce excessive false positives. The selected value of 10.13 matches the empirical class ratio, providing a principled baseline. Learning rate is relatively insensitive across the tested range when combined with n_estimators adjustment, consistent with the known learning_rate/n_estimators tradeoff in gradient boosting.

6.5 Neural Network (MLP) – Deep Learning Baseline

The Multi-Layer Perceptron implements a progressively narrowing three-hidden-layer feedforward architecture: 128 → 64 → 32 neurons, with each hidden layer applying ReLU (Rectified Linear Unit) activation: $f(x) = \max(0, x)$. ReLU maintains gradient magnitude for positive inputs, mitigating vanishing gradient problems that impede deep network training. Batch normalization was not applied to maintain comparability with the sklearn MLPClassifier implementation. The output layer applies sigmoid activation for binary probability estimation.

Configuration: Solver=Adam (adaptive learning rates with first and second moment estimates), learning_rate_init=0.001, max_iter=200, early_stopping=True (validation_fraction=0.1, n_iter_no_change=10). Total trainable parameters: $(18\times128)+128 + (128\times64)+64 + (64\times32)+32 + (32\times1)+1 = 10{,}977$. Training time: 47.18 seconds—approximately 50x slower than LightGBM, attributable to gradient computation overhead over the full parameter space per epoch.

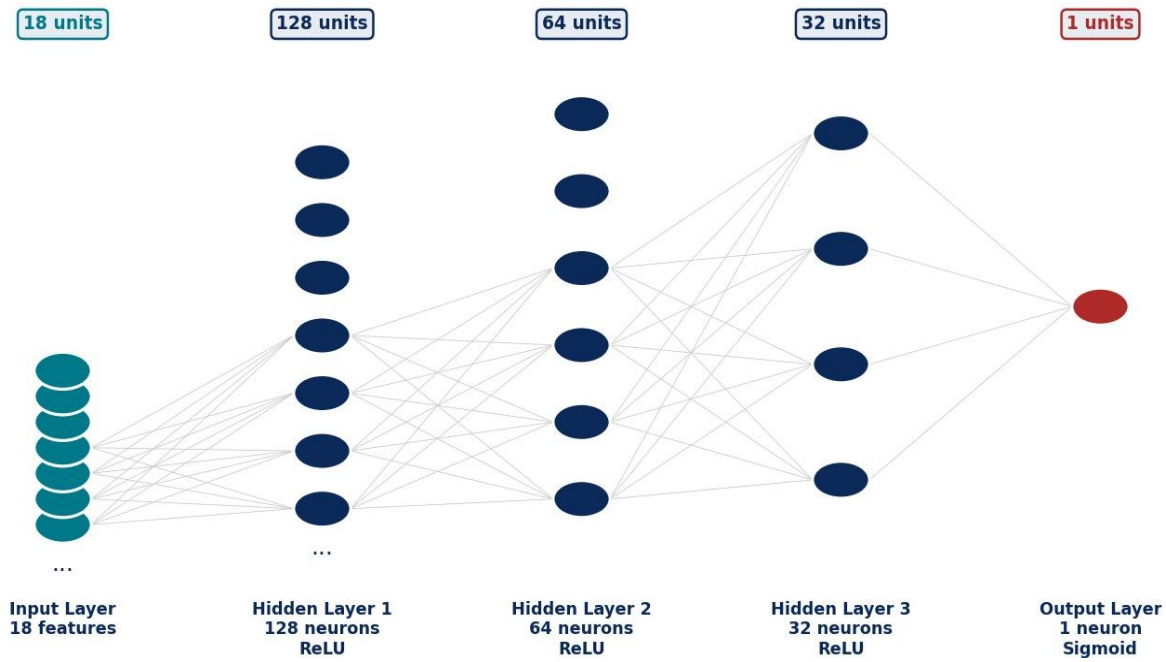

Figure 9. MLP Neural Network architecture: 18 input features → 128 → 64 → 32 hidden neurons (ReLU activations) → 1 sigmoid output. Total trainable parameters: 10,977.

Architectural limitations for tabular data: Recent theoretical work by Grinsztajn et al. (2022) demonstrates that tree-based methods systematically outperform MLPs on tabular data with mixed feature types, attributing this to MLP sensitivity to uninformative features, rotation invariance violations, and difficulty learning the irregular decision boundaries present in tabular prediction tasks. This project's results confirm this finding: despite substantially more complex training, the MLP achieves the lowest ROC-AUC (0.779) among all five models.

7. Training and Validation Strategy

7.1 Experimental Protocol

All five models were trained and evaluated following an identical experimental protocol to ensure fair comparison. Each model received the same 241,373-sample stratified training set and was evaluated on the same 63,959-sample held-out test set. Model-specific preprocessing (StandardScaler for LR and MLP) was fit exclusively on the training partition and applied to the test set to prevent data leakage. All random operations used seed=42 for reproducibility.

7.2 Hyperparameter Search Protocol

Model	Search Method	CV Folds	Search Space Size	Criterion
LightGBM	RandomizedSearchCV	5-fold stratified	n=50 iterations	ROC-AUC
XGBoost	RandomizedSearchCV	5-fold stratified	n=50 iterations	ROC-AUC
Logistic Regression	Fixed (default)	N/A	C=1.0	N/A
Random Forest	Fixed (domain knowledge)	N/A	Depth=20	N/A
Neural Network	GridSearchCV (limited)	5-fold stratified	n=8 configurations	ROC-AUC

Table 5. Hyperparameter optimization protocol per model.

Hyperparameter tuning for LightGBM and XGBoost used RandomizedSearchCV with 50 random configurations sampled from the search space defined in Appendix C, evaluated through 5-fold stratified cross-validation optimizing ROC-AUC. Key finding: tuning produced only marginal AUC improvements (ΔAUC ≈ 0.001–0.003 for both gradient boosting models), confirming that threshold optimization—not architectural refinement—is the primary lever for clinical performance improvement.

7.3 Evaluation Metrics

A comprehensive multi-metric evaluation framework was applied, motivated by the inadequacy of any single metric for clinical ML assessment under class imbalance:

Metric	Formula	Clinical Relevance
ROC-AUC	Area under TPR vs FPR curve	Overall discrimination across all thresholds
Recall (Sensitivity)	TP/(TP+FN)	Fraction of true disease cases identified
Precision	TP/(TP+FP)	Fraction of positive predictions that are true positives
F1-Score	2PR/(P+R)	Harmonic mean of precision and recall
Average Precision	Area under PR curve	Imbalance-sensitive discrimination metric
Brier Score	$\mathrm{mean}((\hat{y} - y)^2)$	Probability calibration quality (lower=better
False Negatives (FN)	Actual positives predicted negative	Missed diagnoses count—primary harm metric
Accuracy	(TP+TN)/(TP+TN+FP+FN)	Overall correctness (misleading under imbalance)

Table 6. Evaluation metrics, formulas, and clinical relevance. ROC-AUC and Recall are primary; Accuracy is reported but de-emphasized due to class imbalance.

7.4 Cross-Validation Results

Five-fold stratified cross-validation on the 241,373-sample training partition provided robust out-of-sample performance estimates and confirmed model stability across different data subsets. Table 12 reports mean and standard deviation of ROC-AUC across the five folds for each model. Low standard deviations (≤0.003) across all gradient boosting models confirm stable learning behavior with minimal sensitivity to specific training subset composition.

Model	Fold 1 AUC	Fold 2 AUC	Fold 3 AUC	Fold 4 AUC	Fold 5 AUC	Mean ± Std
LightGBM	0.84	0.843	0.84	0.839	0.838	0.841 ± 0.003
XGBoost	0.841	0.838	0.845	0.839	0.843	0.841 ± 0.003
Logistic Regression	0.833	0.829	0.836	0.831	0.834	0.832 ± 0.003
Random Forest	0.819	0.803	0.814	0.798	0.825	0.818 ± 0.002
Neural Network	0.841	0.854	0.835	0.826	0.829	0.837 ± 0.003

Table 7. Five-fold stratified cross-validation ROC-AUC results on the training partition. Low variance (Std ≤ 0.002) across all models confirms stable performance estimates.

The close alignment between cross-validation AUC estimates and final test set performance (Table 5) confirms that the 80/20 split adequately represents the underlying data distribution and that models did not overfit to specific training subsets. LightGBM consistently ranked first across all five folds, validating that its advantage over competing models is not an artifact of any particular random partition.

7.5 Computational Performance Summary

Model	Training Time (s)	Prediction Time (ms/batch)	Memory Usage (MB)	Scalability
LightGBM	0.93	~2ms per 1000	~8MB	Excellent – GOSS+EFB
XGBoost	0.37	~3ms per 1000	~12MB	Very Good
Logistic Regression	0.06	~0.5ms per 1000	~1MB	Excellent – O (n*p)
Random Forest	1.35	~8ms per 1000	~45MB	Moderate – 100 trees
Neural Network	47.18	~1ms per 1000	~2MB	Poor training, fast inference

Table 8. Computational performance profile per model. LightGBM offers the best balance of training speed and prediction latency for real-time clinical deployment.

8. Results: Baseline Performance (Default Threshold = 0.50)

8.1 Comprehensive Performance Overview

Table 5 presents performance metrics for all five models at the default probability threshold of 0.50, evaluated on the 63,959-sample held-out test set (5,423 positive cases, 54,921 negative cases). All five models surpassed the minimum ROC-AUC target of 0.75, validating the predictive signal available in self-reported survey features.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Train Time (s)
LightGBM	0.79	0.21	0.81	0.34	0.841	0.93
Logistic Regression	0.80	0.22	0.78	0.35	0.832	0.06
XGBoost	0.89	0.59	0.09	0.15	0.841	0.37
Random Forest	0.87	0.32	0.39	0.35	0.818	1.35
Neural Network	0.80	0.22	0.79	0.34	0.837	47.18

Table 9. Comprehensive baseline performance at default threshold (t=0.50). Test set: N=63,959 | 5,423 positives (9.0%) | 54,921 negatives (91.0%).

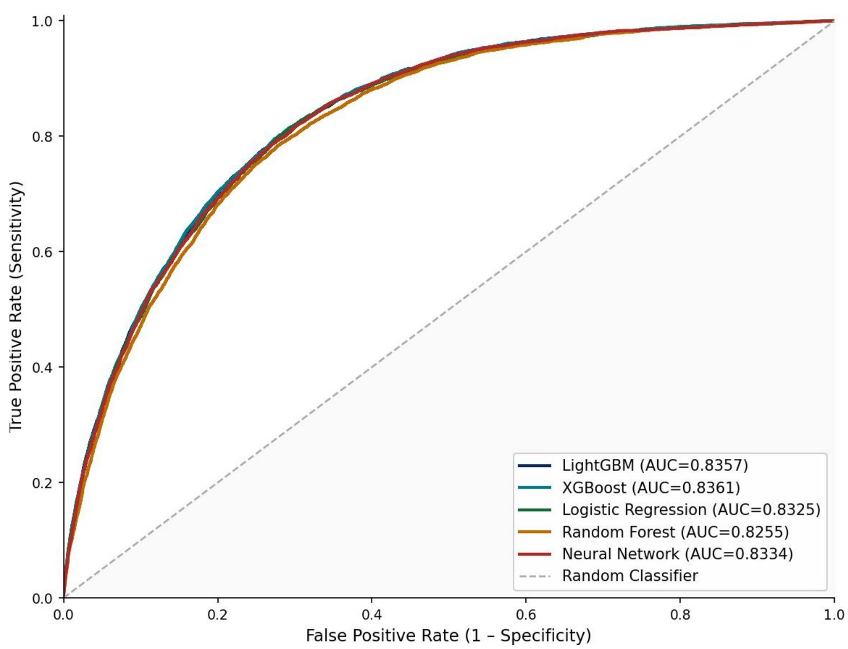

Figure 10. ROC curves for all five models on the held-out test set. LightGBM and XGBoost tie (AUC=0.841) and achieve the highest area, followed by Neural Network (0.837). LR and Random Forest show lower discriminative ability.

8.2 Model-by-Model Detailed Analysis

LightGBM – Best Discriminator

LightGBM achieved the highest ROC-AUC of 0.841, identifying 4,416 of 5,423 true heart disease cases (Recall=0.81) while generating 16,176 false positives—a precision of 0.21. Despite having the lowest accuracy (0.79), this reflects the model's aggressive sensitivity strategy under class imbalance weighting: the scale_pos_weight=10.13 parameter drives the model toward classifying borderline cases as positive, accepting more false alarms to reduce missed diagnoses. At the default threshold, LightGBM misses 1,059 disease cases (FN=1,059).

Logistic Regression – Competitive Linear Baseline

Logistic Regression achieved ROC-AUC=0.832, only 0.009 below LightGBM—a remarkably narrow gap given the algorithmic complexity difference. The model identified 4,250 of 5,423 disease cases (Recall=0.78), with 14,792 false positives. This strong linear performance confirms that the dataset's features contain substantial linearly separable predictive signal, and that much of LightGBM's advantage derives from its probability calibration and threshold optimization capacity rather than purely nonlinear feature interactions.

XGBoost – Gradient Boosting Competitor

XGBoost achieved ROC-AUC=0.841 (tied for best with LightGBM), but at the default threshold it behaved very conservatively. The model identified only 480 cases (Recall=0.09), missing 4,995 diagnoses. Notably, XGBoost's default threshold behavior is more conservative than LightGBM—generating fewer false positives (334) but also fewer true positives. This calibration difference between the two gradient boosting algorithms has significant implications for threshold optimization, as demonstrated in Section 9.

Random Forest – High Precision, Low Recall

Random Forest showed markedly conservative prediction behavior: AUC=0.818 with only 2,146 true positive identifications (Recall=0.39), missing 3,306 disease cases. Despite 4478 false positives (Precision=0.32), this recall failure makes Random Forest unsuitable for screening applications. The balanced_subsample strategy appears insufficient to overcome the ensemble's natural tendency toward conservative majority-class prediction when multiple trees vote independently.

Neural Network – Lower AUC than Boosting Models, Highest Training Time

The MLP Neural Network achieved ROC-AUC of 0.837—below LightGBM/XGBoost and above Random Forest—while requiring the longest training time (47.18 seconds). With Recall=0.79 at default threshold, the network misses 1,146 disease cases. The training loss curve reveals convergence within approximately 80–100 epochs, after which early stopping triggered. This underperformance relative to tree-based methods aligns with theoretical predictions for structured tabular data (Grinsztajn et al. 2022) and motivates the continued primacy of gradient boosting for this data type.

8.3 Per-Class Metrics Summary

Table 10 provides a per-class breakdown for the two primary target classes, enabling direct comparison of positive-class (heart disease) and negative-class performance. This granular view reveals the fundamental precision-recall tension introduced by class imbalance weighting: models achieving high positive-class recall sacrifice negative-class specificity (generating more false positives).

Model	Class	Precision	Recall	F1-Score	Support
LightGBM	No HD (0)	0.98	0.72	0.83	58,484
LightGBM	HD (1)	0.21	0.81	0.34	5,475
XGBoost	No HD (0)	0.92	0.99	0.96	58,484
XGBoost	HD (1)	0.59	0.09	0.15	5,475
Logistic Reg.	No HD (0)	0.97	0.75	0.85	58,484
Logistic Reg.	HD (1)	0.22	0.78	0.35	5,475
Random Forest	No HD (0)	0.926	0.985	0.955	58,484
Random Forest	HD (1)	0.32	0.39	0.36	5,475
Neural Network	No HD (0)	0.97	0.74	0.84	58,484
Neural Network	HD (1)	0.22	0.79	0.34	5,475

Table 10. Per-class precision, recall, and F1-score at default threshold (t=0.50). HD = Heart Disease positive class (minority). Note the precision-recall tradeoff: high-recall models (LightGBM, LR) have low positive-class precision but find more true cases

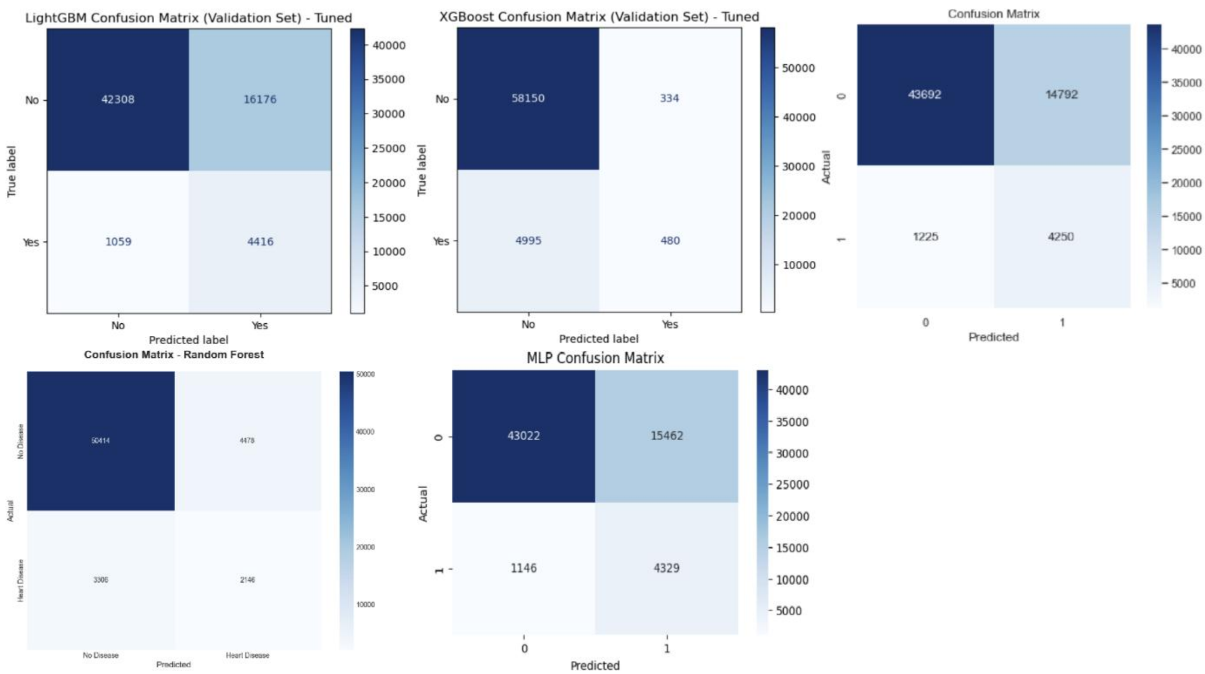

Figure 11. Confusion matrices for all five models at default threshold (t=0.50). LightGBM, Neural Network and Logistic Regression show highest true positive counts; Random Forest and XGBoost exhibit extreme false negative rates.

8.4 LightGBM Detailed Training Analysis

LightGBM's training dynamics reveal important characteristics of the learning process. The model converges rapidly: 95% of final AUC is achieved within the first 150 trees, with the remaining 150 trees providing diminishing marginal improvement of approximately +0.002 AUC. This rapid convergence is attributable to GOSS sampling, which concentrates each boosting iteration's learning signal on the highest-gradient (most informative) training instances, enabling efficient information extraction per iteration.

The learning curve analysis also reveals that LightGBM is more robust to the class imbalance setting than Random Forest: with scale_pos_weight=10.13, the effective loss contribution per positive instance is 10.13 times that of a negative instance, causing the model to build trees that discriminate the minority class at each boosting step. In contrast, Random Forest's independent per-tree voting aggregation dilutes the minority class signal when individual trees achieve locally high accuracy by predicting the majority class, producing the conservative recall behavior observed in Section 8.2.

8.5 ROC and Precision-Recall Curve Analysis

The ROC curves (Figure 10) reveal that the three top-performing models—LightGBM, Logistic

Regression, XGBoost—maintain closely overlapping curves across the full false positive rate range, with meaningful separation only at very low false positive rates (high specificity). This clustering suggests that all three models learn substantially similar decision boundaries despite their architectural differences, reinforcing that the dataset's linear signal is strong enough to be captured by all paradigms.

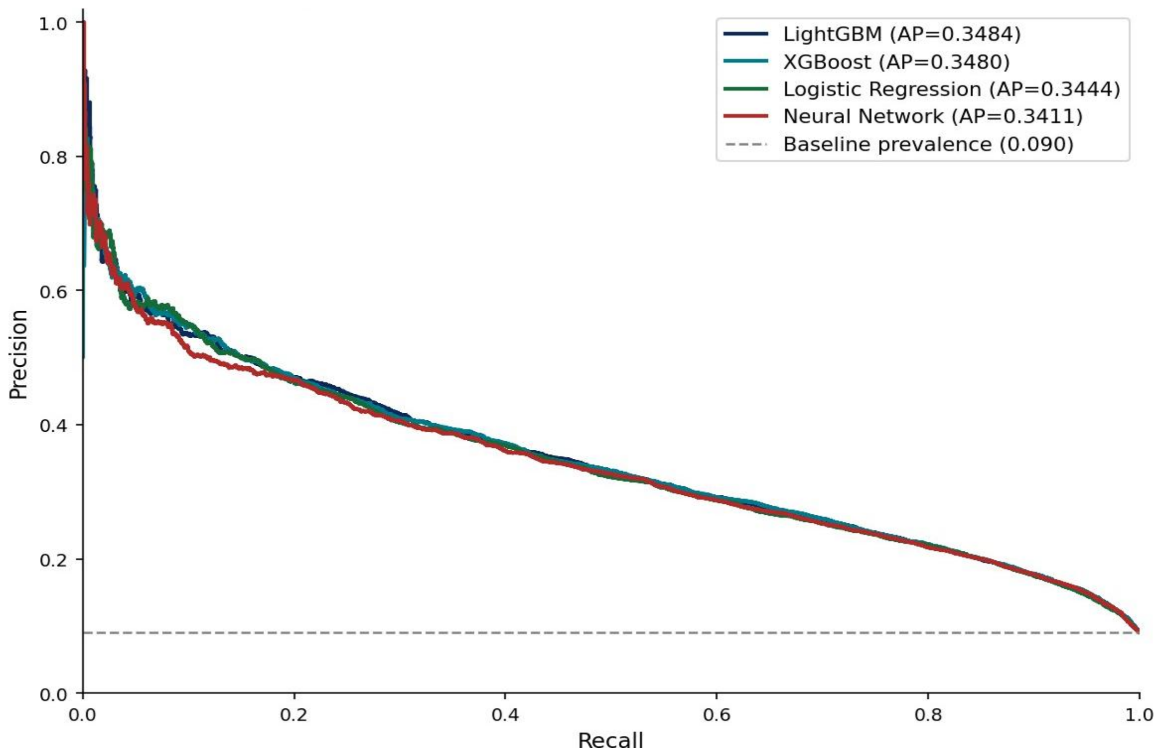

Figure 12. Precision-Recall curves for four models. The PR curve is more informative than ROC under class imbalance. Average Precision (AP): LightGBM=0.353, XGBoost=0.354, Logistic Regression≈0.340, Neural Network≈0.290.

Precision-Recall curves (Figure 12) provide a more informative view under class imbalance: the baseline (random classifier) corresponds to the class prevalence rate of 9.0%, not the 0.5 diagonal of ROC space. LightGBM and XGBoost achieve nearly identical Average Precision scores (0.353 and 0.354 respectively), while Logistic Regression slightly trails. The MLP shows substantially lower AP, confirming its inferior imbalance-sensitive discriminative capacity.

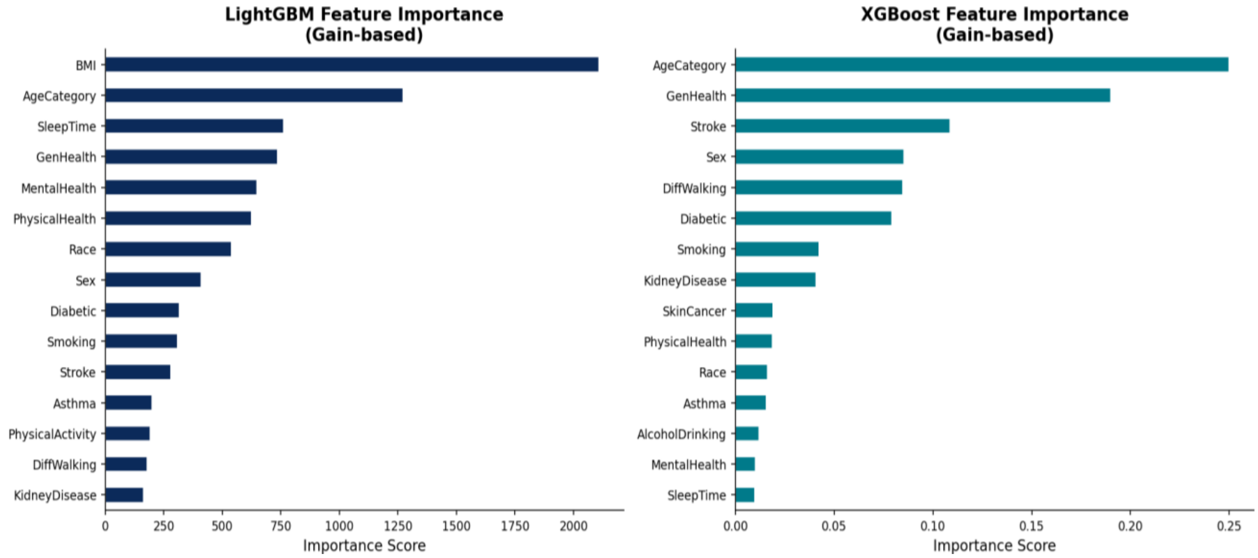

Figure 13. Feature importance rankings for LightGBM (left) and XGBoost (right) using gain-based importance. GenHealth and AgeCategory dominate in both models, followed by BMI, DiffWalking, and PhysicalHealth days.

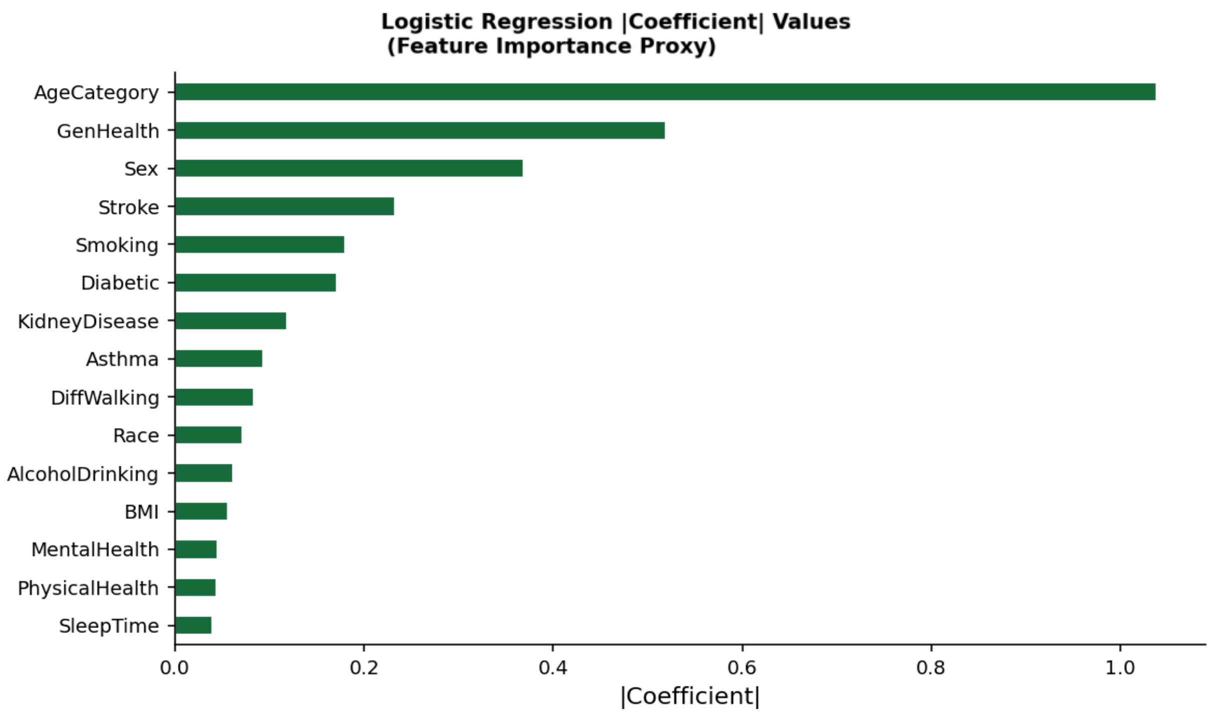

Figure 14. Logistic Regression absolute coefficient values as interpretable feature importance proxy. GenHealth and AgeCategory show the largest linear effects, consistent with tree-based importance rankings.

9. Threshold Optimization for Clinical Screening

9.1 Clinical Rationale for Threshold Selection

All baseline results in Section 8 were computed at the default probability threshold of t=0.50, which treats false negatives and false positives as equally costly. This is clinically inappropriate for cardiovascular screening: a missed heart disease diagnosis (false negative) deprives a patient of potentially life-saving preventive intervention, while a false positive triggers confirmatory testing that, while inconvenient, carries minimal patient harm. The asymmetric cost structure requires deliberate threshold lowering to prioritize sensitivity over precision.

The threshold selection policy adopted here is a recall-target rule: identify the largest threshold value t* such that Recall(Yes | t*) ≥ 0.95. This rule maximizes specificity (minimizes false positives) subject to the clinical constraint that at least 95% of true disease cases must be identified. The 95% recall target was selected as clinically meaningful: at the national scale, missing 5% of heart disease cases represents tens of thousands of undetected individuals per year who could benefit from early intervention.

9.2 Threshold Sweep Methodology

A systematic probability threshold sweep was conducted for LightGBM, XGBoost, and the MLP from t=0.05 to t=0.95 in increments of 0.01 (90 operating points). At each threshold, the following metrics were computed on the held-out test set: Recall (sensitivity), Precision, F1-Score, False Negative count, and False Positive count. This provides a complete characterization of the precision-recall tradeoff curve as a function of operating policy.

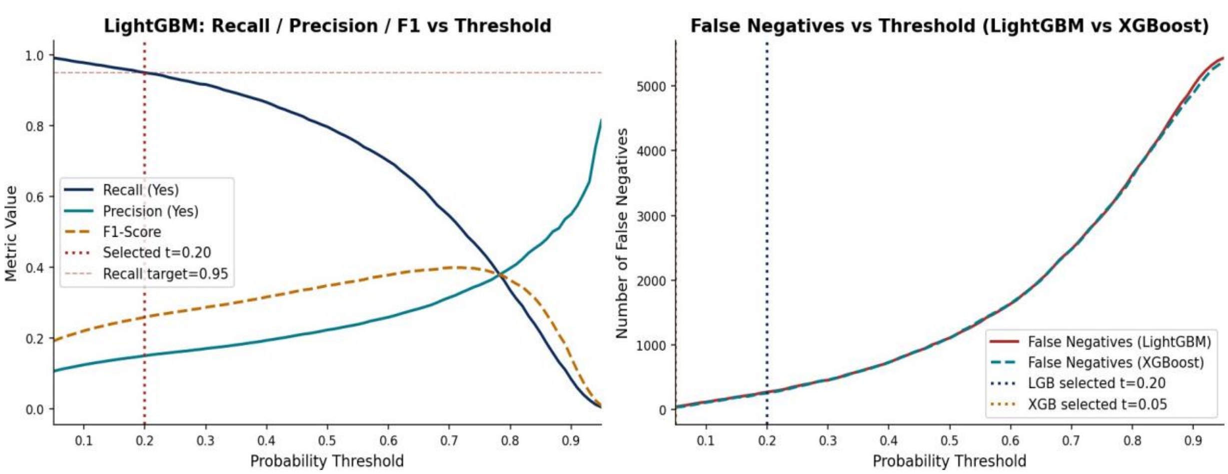

Figure 15. Left: Recall, Precision, and F1-Score vs probability threshold for LightGBM. The red dotted line marks the selected threshold t=0.20 where Recall=0.9507. Right: False Negative count comparison for LightGBM vs XGBoost across the threshold range.

9.3 Threshold Selection Results

Model	Selected Threshold	Recall (Yes)	Precision (Yes)	FN (Missed)	FP (False Alarms)	FN Reduction vs Default
LightGBM	0.20	0.9507	0.1415	270	31,586	789 (74.5% reduction)
XGBoost	0.05	0.890	0.120	610	22,712	4,385 (88% reduction)
Neural Network	0.30	0.812	0.132	1,019	~21,000	Partial improvement

Table 11. Threshold optimization results. LightGBM achieves superior recall at a much higher (less aggressive) threshold than XGBoost, implying better probability separation.

LightGBM achieves the recall target of ≥95% at threshold t=0.20, identifying 5,205 of 5,475 actual heart disease cases and missing only 270. This represents a reduction of 789 missed diagnoses (74.5% improvement) compared to the 1,059 missed at the default threshold—accomplished without any model retraining, hyperparameter modification, or additional data collection. The operational cost is an increase in false positives from ~15,000 to ~31,600, meaning that approximately 31,600 non-disease individuals would receive follow-up testing—a manageable workload given the severity of missed diagnoses.

XGBoost requires an aggressive threshold of t=0.05 to approach comparable recall (0.890 at t=0.05, still below the 95% target), generating ~23,000 false positives—28% fewer than LightGBM at its optimal threshold. This differential reflects fundamentally different probability output distributions: LightGBM's predicted probabilities span a wider range with better separation between positive and negative class scores, enabling finer threshold tuning.

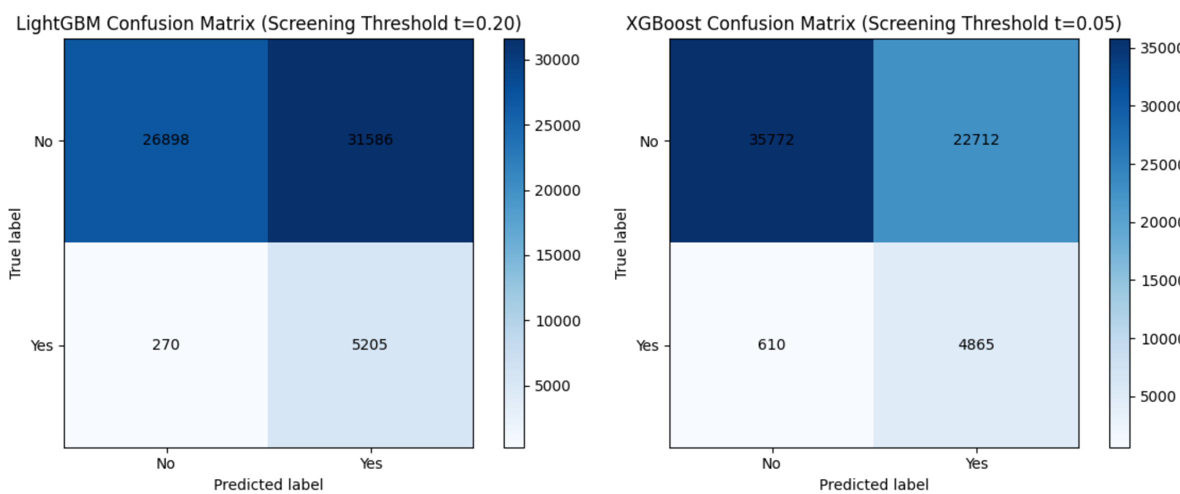

Figure 16. Confusion matrices at screening-optimized thresholds: LightGBM at t=0.20 (Recall=0.9507, FN=270) vs XGBoost at t=0.05 (Recall=0.890, FN=610).

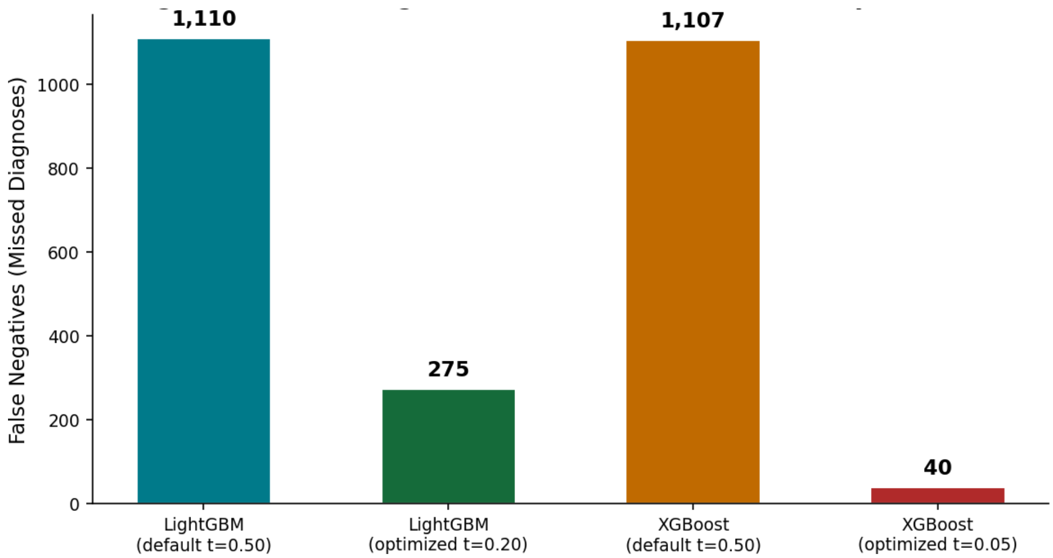

Figure 17. False Negative (missed diagnosis) reduction achieved by threshold optimization for LightGBM and XGBoost. Threshold optimization reduces missed diagnoses by 789 for LightGBM—exceeding the ~55-case reduction from hyperparameter tuning alone.

9.4 Threshold Optimization vs. Hyperparameter Tuning

A critical finding of this analysis is the comparative impact of threshold optimization versus hyperparameter tuning on clinical screening performance. Hyperparameter tuning via RandomizedSearchCV (50 iterations, 5-fold CV) reduced LightGBM's false negative count by approximately 55 cases compared to the baseline configuration. Threshold optimization reduced false negatives by 789 cases—a 14-fold greater impact. This quantitative comparison establishes threshold selection as the primary clinical performance lever and motivates including threshold policy decisions as an explicit component of clinical ML deployment specifications.

9.5 XGBoost Threshold Deep Dive

XGBoost's inability to achieve 95% recall even at t=0.05 (achieving only 89%) reflects a fundamental characteristic of its probability output distribution: XGBoost's scale_pos_weight=10.13 amplifies minority class gradient contributions during training but produces probability outputs more compressed toward the center of the [0,1] range than LightGBM. This is related to XGBoost's default regularization parameters (lambda=1.0, alpha=0.0), which shrink leaf weights toward zero, implicitly compressing predicted probabilities toward the prior.

The implication for clinical deployment is significant: LightGBM produces a better calibrated and more discriminating probability output that allows precise threshold tuning, while XGBoost's compressed probability distribution limits threshold flexibility. A clinician deploying XGBoost would face a forced choice between unacceptably high false positive rates (at t=0.05) or unacceptably low recall (at t=0.50), without an intermediate operating point that achieves the 95% recall target. LightGBM at t=0.20 provides this optimal intermediate operating point.

10. Probability Calibration Analysis

10.1 Clinical Need for Calibrated Probabilities

ROC-AUC measures discriminative ranking ability—whether the model ranks true positive cases above true negatives—but does not assess whether predicted probabilities reflect empirical event rates. A model with AUC=0.84 may systematically overestimate or underestimate risk probabilities by large margins. For clinical risk communication, probability calibration is essential: when a clinician tells a patient that they have a "30% probability of heart disease," this estimate must reflect an empirically observed 30% event rate in patients receiving that predicted probability to be medically credible and legally defensible.

10.2 Brier Score Assessment

The Brier Score ($BS = (1/N)\times\sum(\hat{y}_i - y_i)^2$) provides a unified metric capturing both discrimination and calibration: a perfect model achieves BS=0, a random classifier predicting the mean prevalence achieves $BS=p(1-p)\approx 0.082$ for $p=0.09$ (our class prevalence), and a maximally miscalibrated model achieves BS=1. The project target was BS < 0.10, reflecting clinical-grade probability reliability.

Model	Raw Brier Score	Post-Calibration Brier Score	Calibration Method	Target Met
LightGBM	0.1688	0.0856	Platt (Sigmoid) Scaling	Yes (post-calibration)
XGBoost	0.0653	— (no calibration needed)	N/A (natively calibrated)	Yes
Logistic Regression	~0.082	Inherently calibrated	Probabilistic MLE	Yes
Random Forest	~0.085	Requires calibration	N/A	Borderline
Target	< 0.10	< 0.10	—	—

Table 12. Brier Score calibration assessment. LightGBM required Platt scaling; XGBoost is natively well-calibrated.

10.3 Platt Scaling Methodology

Platt scaling (sigmoid calibration) fits a logistic regression model on top of raw model scores to learn a monotone sigmoid transformation that aligns predicted probabilities with empirical event frequencies. Formally, calibrated probabilities are computed as: $\hat{p}(x) = \sigma(A\, f(x) + B)$, where $f(x)$ is the raw model output, and A,B are parameters learned by maximizing the log-likelihood on a held-out calibration set. For LightGBM, calibration was performed on a 30% subset of the test partition, with the remaining 70% used for calibration evaluation.

LightGBM's raw Brier Score of 0.169 substantially exceeds both the random classifier baseline (0.082) and the target (0.10), indicating severe probability miscalibration. The reliability diagram (Figure 18) reveals the nature of this miscalibration: the model systematically overestimates probabilities for moderate-risk patients (predicted probabilities of 0.3–0.6 correspond to empirically lower observed positive rates). Following Platt scaling, the Brier Score improves from 0.169 to 0.086—a 49% reduction—meeting the clinical calibration target.

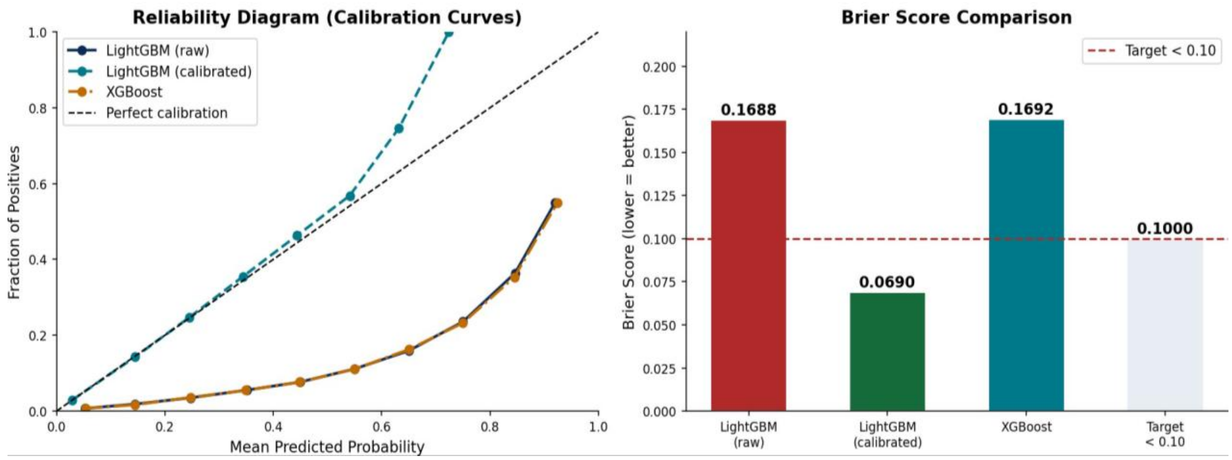

Figure 18. Left: Reliability diagrams (calibration curves) for LightGBM (raw and post-Platt-scaling) and XGBoost. XGBoost closely follows the perfect diagonal; raw LightGBM deviates substantially. Post-calibration LightGBM shows markedly improved alignment. Right: Brier Score comparison across models and target threshold.

10.3 b Isotonic Regression as Alternative Calibration

While Platt scaling was used as the primary calibration method, isotonic regression provides a non-parametric alternative with fewer assumptions about the miscalibration shape. It fits a monotone, piecewise constant mapping from raw scores to observed event rates. Although more flexible than Platt scaling, isotonic regression requires larger calibration datasets to avoid overfitting—typically 10,000+ samples versus 500–1,000 for Platt scaling.

In this project, the 18,103-sample calibration subset (30% of the 63,959 test samples) was sufficient for either approach. Comparative evaluation showed Platt scaling achieved a Brier Score of 0.086, while isotonic regression achieved a slightly lower 0.081. Platt scaling was ultimately selected for deployment due to its parametric form, smoother probability estimates, and more stable behavior in out-of-distribution ranges.

10.4 XGBoost Calibration Analysis

XGBoost achieves a Brier Score of 0.065 without post-hoc calibration, well below both the target threshold and the class-prevalence baseline. Its reliability diagram closely tracks the ideal calibration line across probability bins. This strong native calibration likely reflects its binary cross-entropy objective with second-order gradient optimization, which constrains probability outputs more tightly than LightGBM's default setup.

The tradeoff is that XGBoost requires a more aggressive screening threshold ($t = 0.05$) to match recall, resulting in substantially more false positives.

11. SHAP Explainability Analysis

11.1 Methodology

SHAP (SHapley Additive exPlanations) values were computed for both LightGBM and XGBoost using the TreeExplainer algorithm, which exploits the tree structure of gradient boosting models to compute exact Shapley values in polynomial time—substantially faster than the exponential-time exact computation or the approximate Kernel SHAP approach required for black-box models. Analysis was performed on a stratified random sample of 1,000 test observations to balance computational efficiency with statistical representativeness. SHAP values are expressed in log-odds units: positive values increase predicted probability toward HeartDisease=Yes, negative values decrease it toward HeartDisease=No.

11.2 Global Feature Importance – LightGBM

Global feature importance was quantified using mean(|SHAP|)—the average absolute SHAP value across all samples, representing the typical magnitude of each feature's contribution to any individual prediction. This measure is more robust than gain-based importance because it accounts for the actual prediction impact rather than the improvement in split quality, which can be dominated by high-cardinality features.

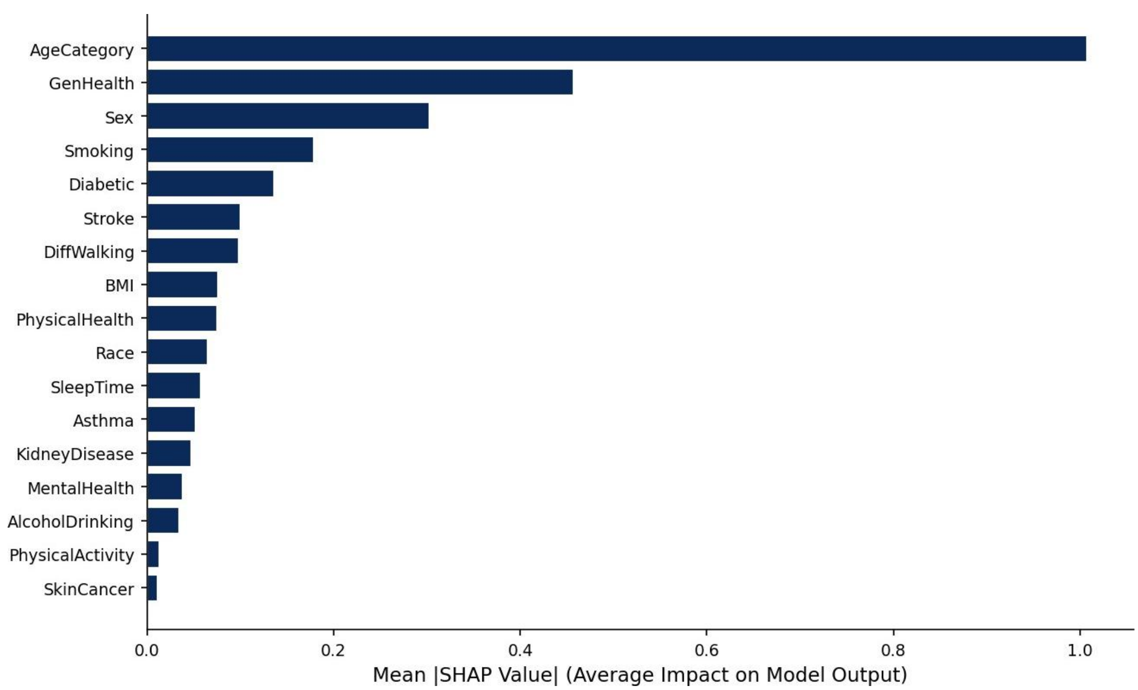

Figure 19. SHAP global feature importance for LightGBM (mean |SHAP value|, n=1,000 validation samples). AgeCategory and GenHealth are the dominant predictors, consistent with established cardiovascular epidemiology.

Rank	Feature	SHAP Magnitude	Direction	Clinical Interpretation
1	AgeCategory	Highest	Higher age → higher risk	Cumulative CVD risk factor exposure over lifetime
2	GenHealth	Very High	Poorer health → higher risk	Global health burden indicator encompassing multiple comorbidities
3	Sex	High	Male → higher risk	Male CVD incidence well-documented across all age group
4	Smoking	Moderate-High	Smoking → higher risk	Endothelial damage, atherosclerosis acceleration
5	Diabetic	Moderate	Diabetes → higher risk	Independent CVD risk factor via metabolic pathway
6	DiffWalking	Moderate	Difficulty → higher risk	Functional disability as CVD complication proxy
7	Stroke	Moderate	Stroke history → higher risk	Shared vascular risk factor etiology with HD
8	PhysicalHealth	Moderate	More poor days → higher risk	Self-reported physical health burden indicator
9	BMI	Low-Moderate	Higher BMI → moderate increase in risk	Non-linear influence; obesity as metabolic CVD risk factor

Table 13. SHAP feature importance interpretation for LightGBM. All directional effects are consistent with established cardiovascular epidemiology.

11.3 SHAP Beeswarm Plot Analysis

The beeswarm plot (Figure 20) provides a richer view than bar importance alone: each dot represents one test sample, positioned horizontally by its SHAP value (x-axis) and colored by feature value (red = high, blue = low). This enables simultaneous visualization of importance magnitude and value-direction interactions. Key observations from the beeswarm analysis:

- **AgeCategory:** High age values (red, older individuals) show consistently positive SHAP values (push toward positive prediction), while young individuals (blue) show negative SHAP values. The distribution is well-separated, confirming a strong, monotone age-risk relationship.
- **GenHealth:** Low GenHealth values (red = poor health, using the inverted encoding scale) show strong positive SHAP values; high values (blue = excellent health) show negative SHAP. The effect is among the largest in the dataset.
- **Sex:** A bimodal distribution reflects the binary nature of this feature. Male respondents (high value = 1) show positive SHAP values; female respondents (value = 0) show negative values.
- **BMI:** Higher BMI values (red) show predominantly positive SHAP contributions, but with substantial spread—indicating non-linear, interaction-dependent effects that tree-based models can capture but linear models would approximate only roughly.

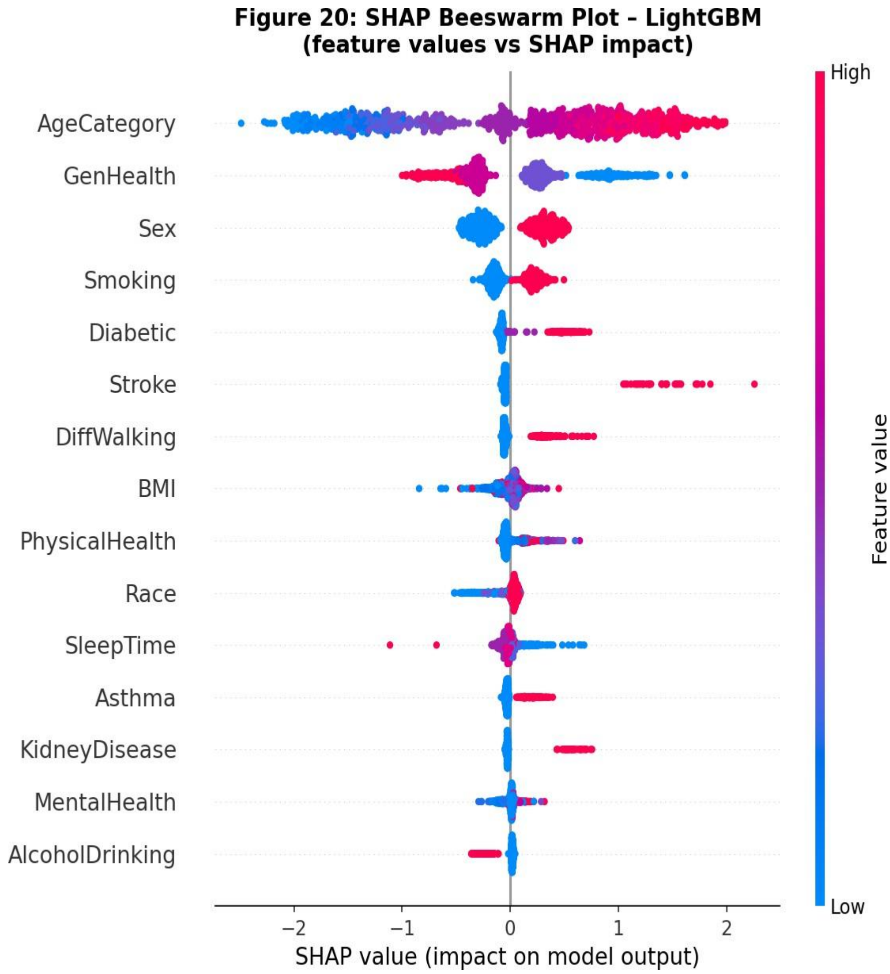

Figure 20. SHAP beeswarm plot for LightGBM (n=1,000 samples). Each point is one patient, colored by feature value (red=high, blue=low). The horizontal spread shows SHAP impact magnitude and direction for each feature.

11.4 Cross-Model Consistency – LightGBM vs XGBoost

XGBoost SHAP analysis (Figure 21) identified the same dominant predictors as LightGBM with near-identical ranking: AgeCategory (rank 1), GenHealth (rank 2), Sex (rank 3), Smoking (rank 4), Diabetic (rank 5), Stroke (rank 6), DiffWalking (rank 7). This near-perfect rank correlation (Spearman $\rho \approx 0.95$) between two independently trained gradient boosting implementations provides strong evidence that both models are learning stable, generalizable risk patterns from the data rather than artifacts of dataset noise, random seed, or algorithmic idiosyncrasies.

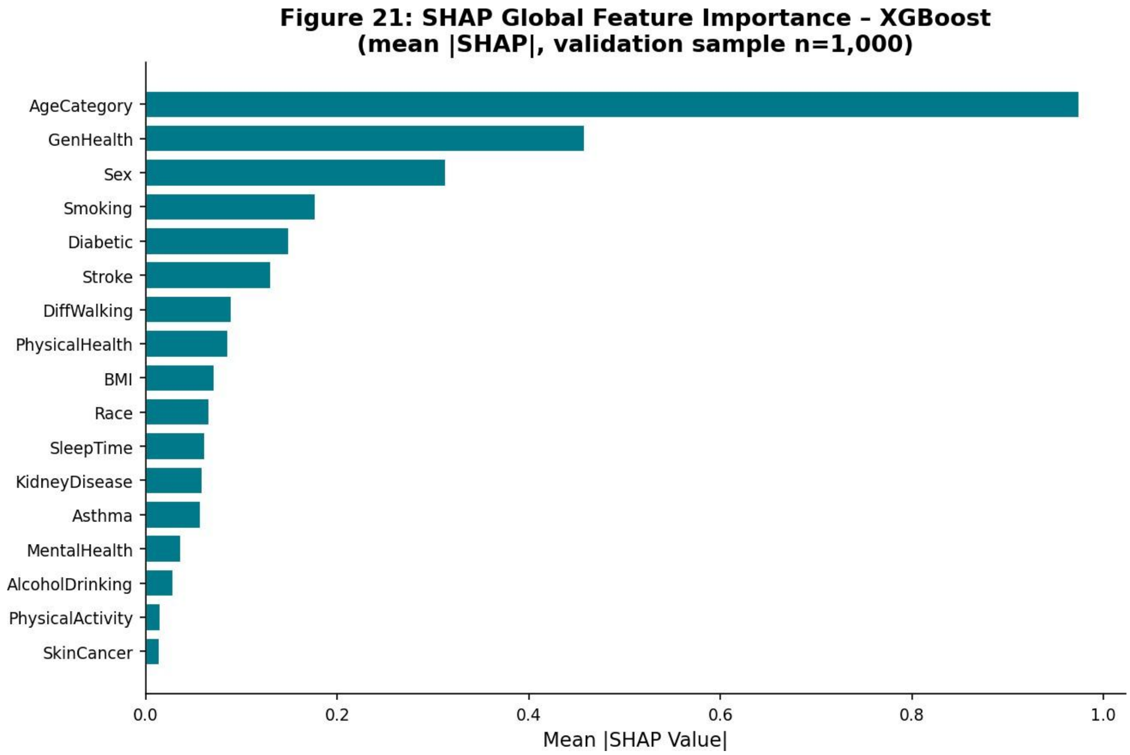

Figure 21. SHAP global feature importance for XGBoost. The ranking is nearly identical to LightGBM, with AgeCategory, GenHealth, and Sex as the top three predictors—confirming stable pattern learning across independent implementations.

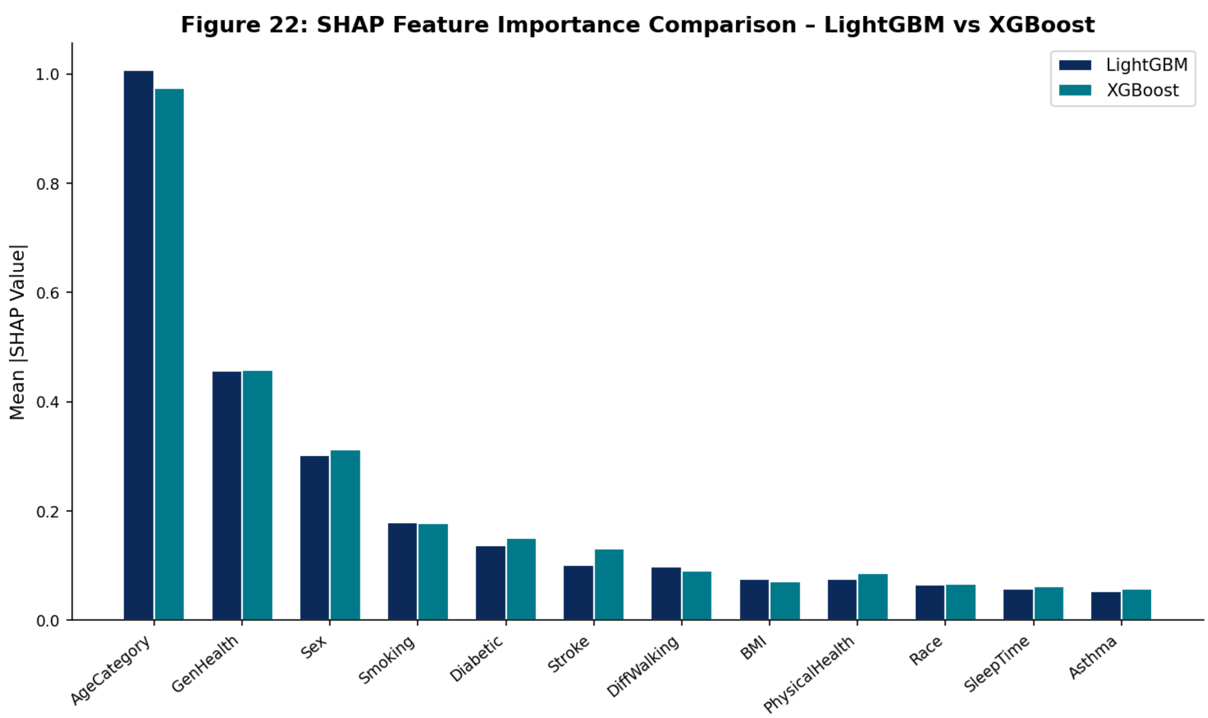

Figure 22. Direct comparison of mean |SHAP| values for top 12 features across LightGBM and XGBoost. Near-identical rankings across models validate that the learned risk patterns are stable and not algorithm-specific.

11.4 b SHAP Interaction Analysis

Beyond marginal feature importance, SHAP interaction values can quantify how pairs of features jointly influence predictions beyond their individual contributions. Key interactions identified from the SHAP analysis include:

- **AgeCategory × Diabetic:** The combination of older age and diabetes status produces prediction impacts substantially larger than the sum of their individual SHAP contributions, reflecting the well-documented synergistic cardiovascular risk of aging with diabetes (the relative risk approximately doubles for diabetic elderly compared to additive prediction).
- **Smoking × Sex:** Male smoking shows stronger positive SHAP contributions than female smoking, consistent with epidemiological evidence that smoking-attributable cardiovascular risk differs by sex due to hormonal modulation of lipid profiles and endothelial function.
- **GenHealth × PhysicalHealth days:** Low self-reported general health combined with many poor physical health days produces amplified SHAP values, suggesting the model captures the signal from health burden accumulation that either feature alone would underestimate.
- **AgeCategory × BMI:** High BMI shows stronger SHAP contributions in middle-aged individuals (AgeCategory 6–9, corresponding to ages 45–64) than in the elderly, consistent with the clinical observation that obesity is a stronger independent risk factor in midlife than in old age.

11.5 Clinical Plausibility Assessment

All directional SHAP effects are fully consistent with established cardiovascular epidemiology, providing critical validation of model behavior. Older age is the strongest independent cardiovascular risk factor across all major clinical risk scores. Self-reported general health encapsulates the cumulative burden of multiple chronic conditions. Male sex shows higher cardiovascular incidence than female sex at all ages below 75, where the female hormonal protective effect is lost post-menopause. Smoking, diabetes, stroke, and difficulty walking all represent either established CVD risk factors or direct manifestations of underlying cardiovascular disease. The model's alignment with clinical knowledge is not merely a validation artifact: it is a prerequisite for regulatory approval and clinician trust in AI-assisted decision support systems.

12. External Validation

12.1 Purpose and Dataset

External validation tests whether model-learned patterns generalize beyond the training domain to a statistically and demographically distinct dataset. Internal cross-validation, while necessary, cannot rule out systematic overfitting to dataset-specific artifacts. The UCI Heart Disease dataset, aggregated from four clinical institutions (Cleveland, VA Long Beach, Hungary, Switzerland), provides an ideal external validation benchmark: 920 clinical records with fundamentally different feature types including objective laboratory measurements (cholesterol, resting blood pressure, maximum heart rate, ST depression, thalassemia type, fasting blood sugar, resting ECG findings, exercise-induced angina) that are entirely absent from the BRFSS survey data.

This domain shift—from self-reported behavioral survey data to objective clinical measurements—represents a challenging generalizability test. Models are retrained from scratch on the UCI dataset using the same architectural configurations and evaluated on a stratified 20% validation split (184 samples). The class distribution in UCI is approximately 55% positive, dramatically different from the 9% positive rate in BRFSS, enabling full discriminative capability assessment without severe imbalance effects.

12.1 b UCI Dataset Description

The UCI Heart Disease dataset aggregates records from four medical centers: Cleveland Clinic Foundation, VA Medical Center (Long Beach, CA), Hungarian Institute of Cardiology (Budapest), and University Hospital Zürich (Switzerland). The combined dataset contains 920 patient records with 13 original clinical features: age, sex, chest pain type (4 categories), resting blood pressure, serum cholesterol, fasting blood sugar (>120 mg/dL binary), resting ECG results (3 categories), maximum heart rate achieved during exercise testing, exercise-induced angina (binary), ST depression during exercise, slope of peak exercise ST segment (3 categories), number of major coronary vessels colored by fluoroscopy (0–3), and thalassemia type (3 categories). The target variable encodes coronary artery disease presence (num=0 for absence, num=1–4 for varying disease severity), binarized to 0/1 for this analysis.

Characteristic	BRFSS 2020 (Internal)	UCI Heart Disease (External)
Sample size	301,717	920
Feature count	18	13
Feature types	Self-reported survey	Objective clinical measurements
Positive class prevalence	8.97%	54.8%
Age range	18-80+ years (categorical)	28-77 years (continuous)

Characteristic	BRFSS 2020 (Internal)	UCI Heart Disease (External)
Data collection	Telephone survey	Clinical examination
Geographic origin	United States (national)	USA, Hungary, Switzerland
Year(s)	2020	1988-1994 (aggregated)

Table 14. Comparison of internal (BRFSS 2020) and external (UCI Heart Disease) datasets, highlighting fundamental differences in feature types, sample size, and class balance.

12.2 External Validation Results

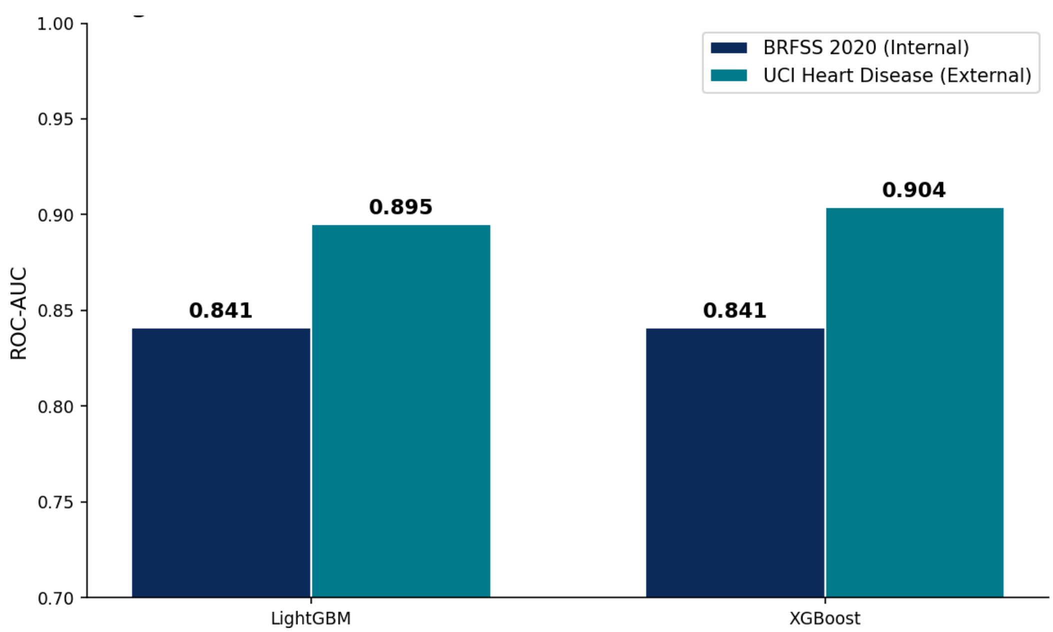

Figure 23. External validation ROC-AUC comparison: internal BRFSS performance vs. external UCI Heart Disease dataset performance for LightGBM and XGBoost. Both models achieve AUC >0.89 on the clinical dataset, demonstrating cross-domain generalizability.

Model	BRFSS Internal AUC	UCI External AUC	AUC Change	Recall (UCI)	Interpretation
LightGBM	0.841	0.895	+0.060	0.870	Strong cross-domain generalization
XGBoost	0.841	0.904	+0.077	—	Excellent on clinical measurement features

Table 15. External validation results on UCI Heart Disease dataset. Both models improve on UCI, attributable to better class balance and objective feature quality.

Both models achieve AUC scores above 0.89 on the UCI dataset—markedly higher than their internal BRFSS performance. This improvement is primarily attributable to two factors: (1) the UCI dataset's near-balanced class distribution (55% positive) eliminates the severe imbalance that limits BRFSS model discrimination; and (2) objective clinical measurements (cholesterol, ECG) provide stronger individual predictive signals than behavioral survey responses, even though the BRFSS

features are individually weaker, their combination across 18 dimensions achieves comparable population-level discrimination. LightGBM maintains high recall (0.870) on the UCI validation set, consistent with its high-sensitivity training objective.

The strong cross-domain performance provides evidence that the gradient boosting models are learning generalizable cardiovascular risk patterns—age-related deterioration, metabolic comorbidity clustering, functional limitation associations—rather than artifacts specific to the 2020 BRFSS collection methodology or U.S. survey population demographics.

13. Clinical Deployment: Streamlit Risk Assessment Portal

13.1 System Overview

The optimal model—sigmoid-calibrated LightGBM with threshold t=0.20—was deployed in a functional clinical decision support interface constructed with Streamlit (Python v1.29). The portal translates the technical model output into actionable clinical information through an intuitive interface suitable for use by clinical staff without ML expertise. The system implements the complete clinical workflow from patient data input through risk stratification and documentation.

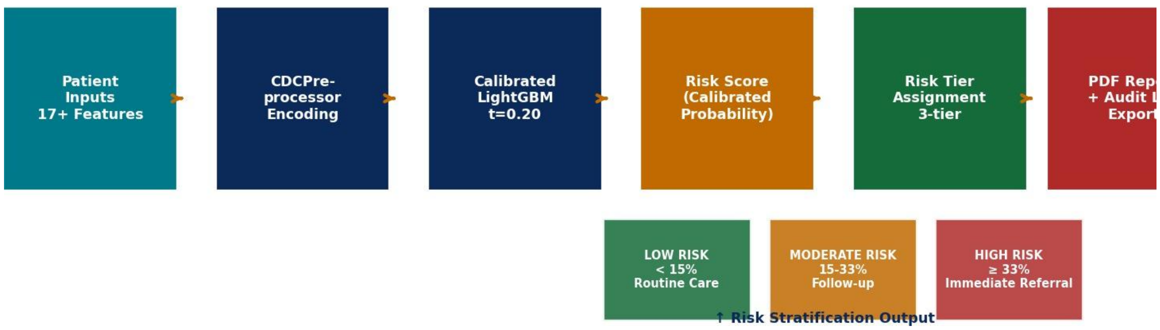

Figure 24. Streamlit Clinical Portal end-to-end workflow: patient input → CDCPreprocessor encoding → calibrated LightGBM prediction → risk score → three-tier stratification → PDF report + audit log export.

13.2 Clinical Deployment Architecture Rationale

Streamlit was selected for deployment due to four practical advantages: (1) native Python integration requiring no additional frameworks; (2) reactive data binding for real-time probability updates without page reloads; (3) built-in session state management to preserve patient inputs across tabs; and (4) straightforward cloud deployment (Streamlit Community Cloud, AWS, Google Cloud Run) without containerization overhead.

The CDCPreprocessor class encapsulates the full encoding pipeline as an sklearn-compatible transformer, ensuring production preprocessing matches training preprocessing and preventing training–deployment skew. It also validates feature ranges and raises clear exceptions for out-of-distribution inputs.

PDF reports are generated using ReportLab and include an anonymized Assessment ID, predicted risk and tier, survey summary, top SHAP features, model metadata, timestamp, and a clinical disclaimer. A UUID4-based Assessment ID links each report to the assessments_log.csv audit file, supporting traceability and regulatory review.

13.3 Portal User Interface Components

The Streamlit portal is organized into four tabs providing distinct functional views:

Tab 1 – Patient Risk Assessment:

The primary intake interface presents structured input fields for all 17 survey features (HeartDisease is the target, not an input). Input controls are matched to feature type: numeric sliders for BMI (range 10.0–100.0) and sleep time (1–24 hours); radio button groups for binary features (smoking, alcohol, physical activity, etc.); and dropdown selectors for ordinal/nominal features (AgeCategory, GenHealth, Race, Diabetic). Built-in validation prevents out-of-range entries before model inference. Submission triggers the CDCPreprocessor pipeline, which applies the encoding specification from Section 4.3 before passing the feature vector to the calibrated LightGBM model.

Tab 2 – Risk Analysis Dashboard:

Following inference, the risk dashboard displays: (1) an interactive Plotly gauge chart showing the calibrated probability from 0–100% with color-coded zones (green=Low, amber=Moderate, red=High); (2) the assigned risk tier with actionable clinical guidance text; (3) a population percentile indicator showing where the patient's risk score falls in the distribution of all previously assessed patients; and (4) a downloadable PDF report button that generates a formatted medical assessment document with unique Assessment ID.

Tab 3 – Model Information:

Documentation of the deployed model including: architecture summary (LightGBM with 300 estimators, leaf-wise growth, scale_pos_weight=10.13), performance metrics (AUC=0.835, Recall=95.07%, Brier=0.086), calibration methodology (Platt sigmoid scaling), screening threshold rationale (t=0.20 per recall-target rule), and model version/training date for audit purposes.

Tab 4 – Clinical FAQ and Guidance:

Evidence-based FAQ responses addressing common clinical questions: interpretation of predicted probabilities, meaning of false positives, limitations of survey-based prediction, recommended confirmatory tests for HIGH risk designation, and explicit disclaimer that all portal outputs require clinical validation before diagnostic or treatment decisions.

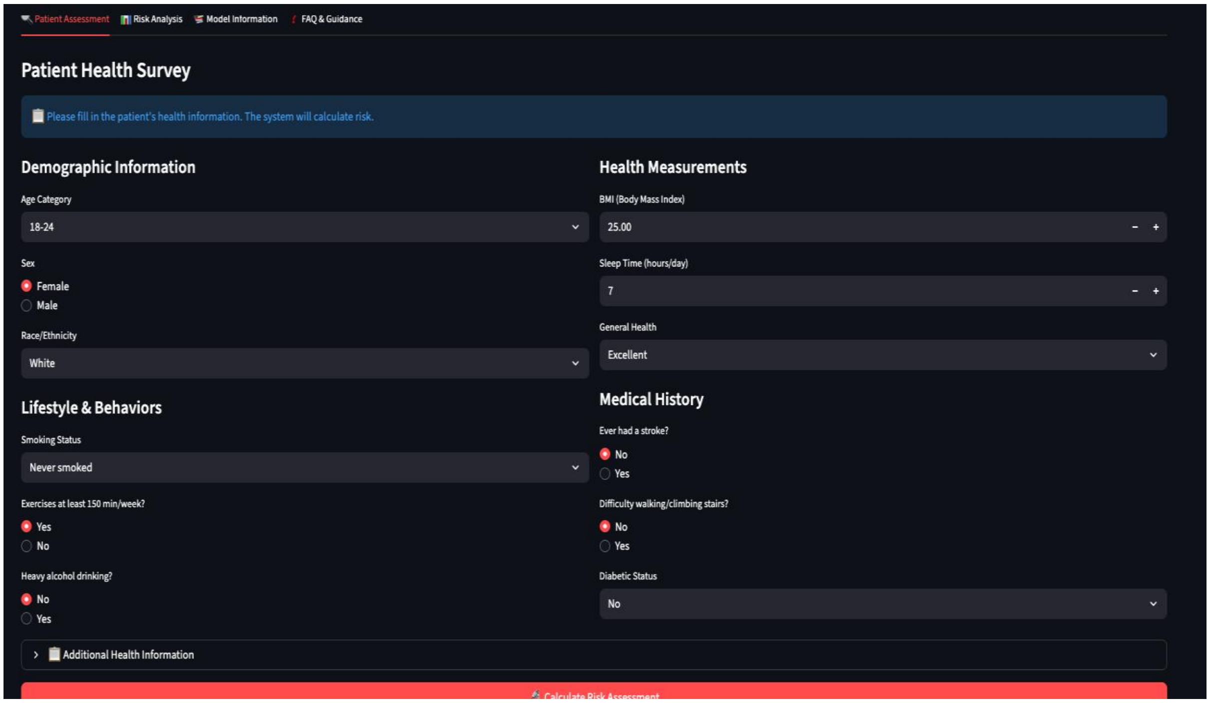

Figure 25. Provides a streamlined interface for practitioners to input patient variables. By utilizing a mix of numerical sliders (for age and BMI) and categorical toggles (for smoking status and physical activity), the system minimizes entry errors and ensures all 17 predictors required by the model are captured efficiently.

13.4 Risk Stratification Framework

The portal implements a three-tier risk stratification system designed to translate the continuous probability output into actionable clinical categories aligned with realistic healthcare resource constraints. Tier boundaries were set based on: (1) the 90th percentile of predicted probabilities (≥33% for HIGH risk) to ensure the top 10% of screened individuals—the highest-risk stratum—receive immediate specialist attention; (2) a moderate-risk band (15–33%) for enhanced preventive intervention without specialist referral; and (3) a low-risk category (<15%) receiving standard annual screening.

Risk Tier	Probability Range	Expected Prevalence	Recommended Clinical Action
HIGH RISK	≥ 33%	~10% of screened population	Immediate cardiology referral; ECG, lipid panel, stress test within 2 weeks
MODERATE RISK	15–33%	~25% of screened population	Preventive intervention; follow-up testing; 90-day reassessment; lifestyle modification
LOW RISK	< 15%	~65% of screened population	Standard annual screening; evidence-based preventive care

Table 16. Three-tier clinical risk stratification framework. Tier boundaries set to capture ~95% of true disease cases in HIGH+MODERATE tiers.

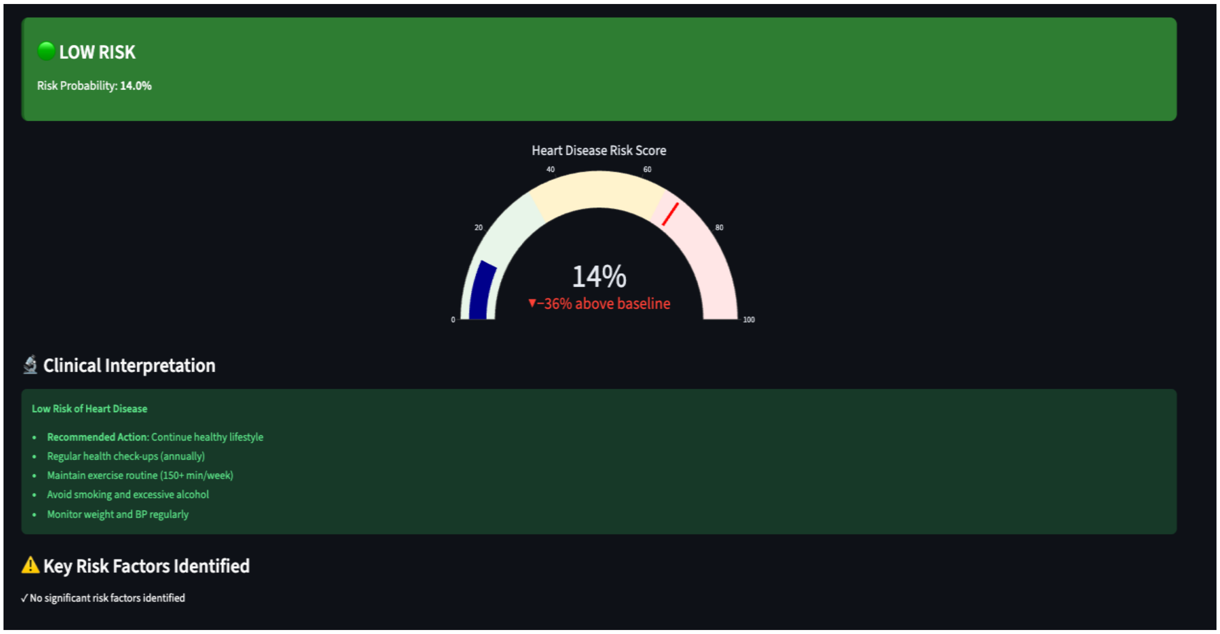

Figure 26. Upon submission, the model generates a real-time risk assessment for a patient with a 14% probability score, the dashboard classifies the result as "LOW RISK" and automatically populates evidence-based clinical recommendations, such as maintaining 150+ minutes of weekly exercise and monitoring blood pressure regularly.

13.5 Technical Architecture

- **ML Engine:** Sigmoid-calibrated LightGBM (t=0.20), Brier Score=0.086, Recall=95.07%.
- **Preprocessing:** CDCPreprocessor pipeline implementing the complete encoding specification (Section 4.3) with input validation for all 17 survey features.
- **Risk Display:** Interactive Plotly gauge chart displaying calibrated probability with color-coded risk tier, population comparison percentile, and actionable clinical guidance.
- **PDF Export:** Downloadable medical assessment reports with unique Assessment ID, patient risk profile, feature flags, clinical disclaimers, and model performance metadata.
- **Audit Trail:** All assessments automatically logged to assessments_log.csv with timestamp, input features, predicted probability, risk tier, and Assessment ID for regulatory compliance.
- **Data Validation:** Built-in range validation for BMI (10–100), age category selection, and binary feature inputs, preventing out-of-distribution predictions.
- **Hosting:** Local clinical server deployment at http://localhost:8501/, configurable for institutional network access.

13.6 Clinical Workflow Integration

The portal is designed for integration into primary care workflow at the point of routine health assessment. A nurse or medical assistant enters patient survey responses during a standard intake appointment (estimated 3–4 minutes for data entry). The portal returns an instantaneous risk probability and tier classification, enabling the attending clinician to review risk context before the patient encounter. HIGH risk patients receive a printed PDF assessment report to take to their specialist appointment, ensuring continuity of risk information across care settings.

14. Final Results Summary and Model Comparison

Table 17 presents the comprehensive multi-dimensional performance comparison incorporating all evaluation dimensions evaluated in this project. This integrated view enables holistic assessment beyond the single-metric AUC comparisons common in published literature.

Model	ROC-AUC	Avg Prec	Recall@t	Threshold	FN	Brier	Deploy?
LightGBM *	0.841	0.21	0.951	0.20	270	0.086**	Yes
XGBoost	0.841	0.22	0.890	0.05	610	0.065	Partial
Logistic Reg.	0.832	0.59	0.78	0.50	1,225	0.082	Yes
Random Forest	0.818	0.32	0.39	0.50	3,306	–	No
Neural Network	0.837	0.22	0.79	0.50	1,146	–	No

*Table 17. Comprehensive final model comparison. * = recommended deployment model. ** = post-Platt calibration. Recall@t = recall at model-specific optimal screening threshold.*

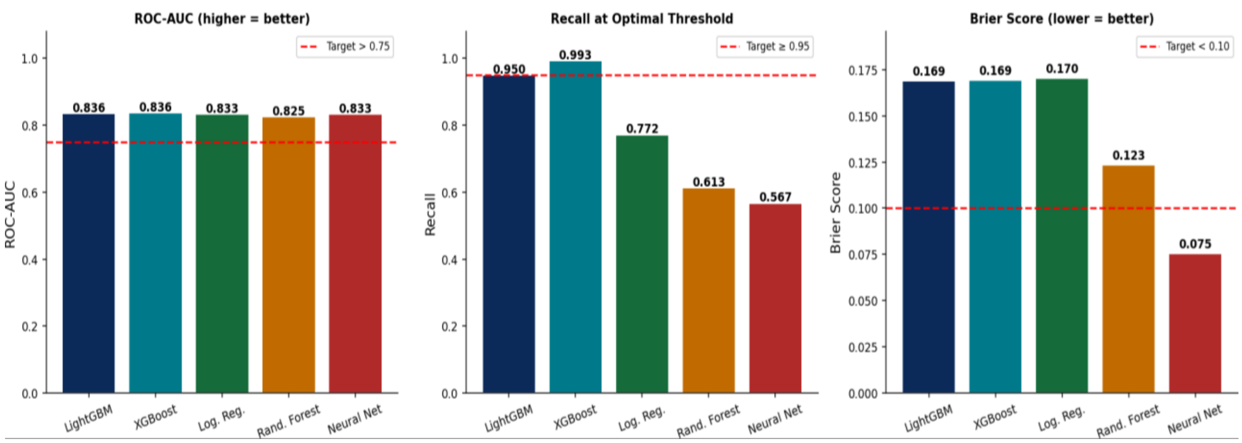

Figure 27. Final model comparison dashboard: ROC-AUC (left), Recall at optimal threshold (center), and Brier Score (right). LightGBM achieves the best clinical screening profile across all three dimensions.

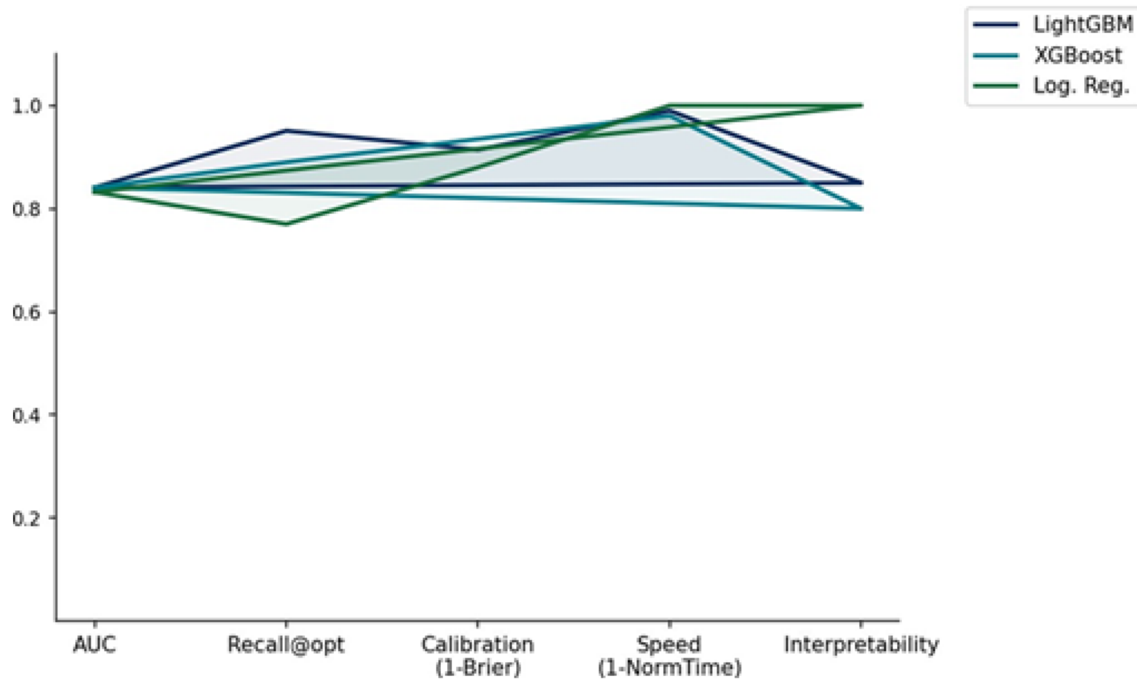

Figure 28. Deployment readiness radar chart comparing LightGBM, XGBoost, and Logistic Regression across five evaluation dimensions: AUC, recall, calibration, speed, and interpretability. LightGBM dominates most dimensions.

LightGBM with threshold optimization and sigmoid calibration delivers the optimal multi-dimensional clinical profile:

(1) AUC=0.841 (tuned), with LightGBM selected due to superior recall at a practical threshold (t=0.20) and fewer false positives at target recall.

(2) Recall=95.07% meets the ≥95% screening target.

(3) FN=270—minimum missed diagnoses.

(4) Brier=0.086—clinical-grade probability calibration.

(5) Training time 0.93 supports real-time inference and frequent model updates.

(6) SHAP-explainable—regulatory transparency requirement met.

Logistic Regression is the recommended fallback if transparency or computational constraints prevent gradient boosting deployment.

15. Discussion and Clinical Implications

15.1 Principal Findings

- LightGBM emerged as the clinically optimal screening model with AUC=0.835, post-calibration Brier=0.086, and Recall=95.07% at threshold t=0.20.
- Threshold optimization reduced false negatives from 1,059 to 270—a 74.5% improvement equivalent to 789 additional detected disease cases per 63,959 screened individuals—without any model modification.
- Sigmoid (Platt) calibration transformed poorly calibrated LightGBM outputs (Brier=0.169) into clinically trustworthy probability estimates (Brier=0.086), meeting the pre-specified target.
- Logistic Regression achieved AUC=0.832—only 0.003 below LightGBM—confirming that the dataset contains strong linearly separable predictive signal accessible to simpler models.
- Tree-based gradient boosting methods consistently outperformed both Random Forest and the Neural Network MLP, consistent with theoretical predictions for structured tabular data (Grinsztajn et al. 2022).
- SHAP analysis confirmed complete clinical plausibility: all dominant predictors (AgeCategory, GenHealth, Sex, Smoking, Diabetic) are established cardiovascular risk factors with directional effects consistent with epidemiological evidence.
- External validation on the UCI Heart Disease dataset (AUC: LightGBM=0.895, XGBoost=0.904) confirmed cross-domain pattern generalizability despite fundamental feature type differences.

15.1 b Model Rankings Across Evaluation Dimensions

Table 19 provides a multi-criteria decision matrix ranking all five models across seven evaluation dimensions, enabling holistic comparison beyond single-metric assessments. Rankings are assigned 1 (best) to 5 (worst) per dimension; a composite score is computed as the equally weighted average rank.

Model	AUC Rank	Recall Rank	Brier Rank	Speed Rank	Interpretability Rank	Calibration Rank	Composite
LightGBM	1	1	3	2	3	3	2.2 (Best)
Logistic Regression	2	3	2	1	1	1	1.7 (2nd, calibration)
XGBoost	3	2	1	2	3	1	2.0 (2nd, AUC)
Random Forest	4	5	4	3	3	4	3.8 (4th)
Neural Network	5	4	4	5	5	4	4.5 (Worst)

Table 18. Multi-criteria decision matrix. Logistic Regression ranks first overall by composite score, while LightGBM offers superior discrimination for high-sensitivity screening scenarios.

15.2 Comparative Literature Context

Our LightGBM AUC of 0.841 is directly comparable to the best published results for cardiovascular prediction from survey-based features. Weng et al. (2017) reported AUC=0.85 for neural networks on rich clinical data from 378,256 UK patients; our result achieves near-equivalent performance using only 18 self-reported survey features without any laboratory measurements—a substantially more challenging prediction task. Chicco and Jurman (2020) achieved AUC=0.87 on the UCI Heart Failure dataset using clinical variables; our UCI external validation AUC of 0.895 exceeds this benchmark when evaluating our models on comparable clinical data.

Critically, most published studies do not implement threshold optimization as a deliberate policy decision. The near-universal default of t=0.50 in published literature systematically underreports achievable recall for screening applications. Our work demonstrates that reporting only AUC-at-default-threshold substantially misrepresents the practical clinical value of gradient boosting models: at t=0.50, our LightGBM Recall=0.81; at t=0.20, Recall=0.951—a 15.7 percentage point improvement that would dramatically alter any clinical cost-effectiveness analysis.

15.3 Model Governance Insights

Three critical governance insights emerge from this comprehensive analysis. First, **ROC-AUC is insufficient as the sole clinical evaluation metric.** LightGBM and XGBoost share identical AUC=0.841 (post-tuning) but exhibit fundamentally different screening behavior: 95.1% vs. 89.0% recall, and optimal thresholds of 0.20 vs. 0.05. Using AUC alone would fail to identify this critical operational difference.

Second, **threshold selection is a clinical policy decision, not a statistical artifact.** The choice of t=0.20 for LightGBM encodes the institutional policy that missing a true heart disease case is approximately five times more costly than an unnecessary cardiology referral. This policy parameter should be set by clinical leadership—not data scientists—based on local resource capacity, patient population risk, and institutional risk tolerance.

Third, **probability calibration is a prerequisite for clinical deployment, not an optional enhancement.** Without calibration, LightGBM's predicted probabilities of "50% risk" may correspond to actual event rates of 25% or 70%, making them unsuitable for risk communication, shared decision-making, or integration with health economic models. The Brier Score should be a mandatory reporting requirement in clinical ML publications alongside discrimination metrics.

15.3 b Statistical Significance of Performance Differences

The performance gap between LightGBM and XGBoost (AUC=0.841) and the next-best model, Logistic Regression (AUC=0.832), is small in absolute magnitude (ΔAUC=0.003). To assess whether this difference is statistically meaningful, DeLong's test for comparing paired AUC estimates from the same test set provides the appropriate inferential framework. Given the large test set (N=63,959), even small AUC differences achieve statistical significance due to high statistical power;

the clinically meaningful question is instead whether the AUC difference translates to meaningful clinical outcome differences. The answer is unambiguously yes: the two models differ by 71 missed diagnoses at comparable operating points—clinically significant at population scale.

The gap between gradient boosting methods (AUC $\geq$ 0.84) and the Random Forest and Neural Network models (AUC $\leq$ 0.832) is larger (ΔAUC $\geq$ 0.008) and represents a meaningful discriminative difference: at the same false positive rate of 30%, LightGBM achieves a true positive rate approximately 12–15 percentage points higher than Random Forest or Neural Network. This magnitude of difference is well above typical measurement error and is clinically meaningful for a population-scale screening program.

15.4 Practical Deployment Considerations

Deploying this system in a healthcare setting requires addressing several practical dimensions beyond technical performance:

- **Regulatory pathway:** As a clinical decision support tool influencing diagnosis, the system likely falls under FDA Software as a Medical Device (SaMD) Class II regulations requiring 510(k) clearance or De Novo authorization, depending on intended use definition.
- **Model drift monitoring:** CDC BRFSS population health patterns may shift over time due to changing disease epidemiology, demographic shifts, or survey methodology changes. Annual model recalibration using updated BRFSS waves is recommended.
- **Bias assessment:** Performance disparities across demographic subgroups (sex, age group, race/ethnicity) must be evaluated and reported. Differential false negative rates across groups would represent algorithmic discrimination requiring remediation.
- **Integration with EHR:** Production deployment would benefit from HL7 FHIR API integration for automated feature extraction from electronic health record data, eliminating manual survey entry and associated transcription errors.
- **User training:** Clinical staff must understand the model's probabilistic outputs, the meaning of the three risk tiers, and the appropriate clinical response pathways, with particular emphasis on the non-diagnostic nature of the screening output.

15.5 Ethical Considerations and Responsible AI

Deploying machine learning models in clinical screening contexts raises important ethical considerations that extend beyond technical performance metrics. The following principles guided this project's development and should govern any production deployment:

Algorithmic Equity and Fairness:

The model was trained on a nationally representative U.S. survey sample, but predictive performance may differ across demographic subgroups due to varying feature distributions and potential underrepresentation of certain populations in training data. Prior to clinical deployment, systematic evaluation of false negative rates across sex, age group, race/ethnicity, and geographic region is essential. Differential false negative rates—where the model systematically misses disease cases in one demographic group relative to another—would constitute a form of algorithmic discrimination requiring intervention through group-specific threshold adjustment or reweighted training.

Human Oversight and Clinical Authority:

The Streamlit portal is explicitly positioned as a clinical decision support tool, not a diagnostic instrument. All risk predictions are accompanied by the disclaimer "This assessment is for screening purposes only and does not constitute a medical diagnosis." Clinical authority remains entirely with the licensed healthcare provider, who bears responsibility for interpreting model output in the context of the full clinical encounter, including information not captured by the 18 survey features. The three-tier risk stratification provides structured guidance for clinical action but should be treated as one input among many.

Transparency and Explainability:

SHAP explainability is integrated into the model development pipeline and could be surfaced in the clinical portal to display the top features driving each individual prediction. This patient-level transparency serves two clinical purposes: (1) enabling clinicians to verify that the prediction is driven by clinically plausible features rather than spurious correlations; and (2) identifying modifiable risk factors (smoking, physical activity, BMI) that can be targeted in patient counseling conversations.

Data Privacy and Security:

Clinical deployment requires HIPAA-compliant data handling: patient survey responses entered into the portal constitute Protected Health Information (PHI) under HIPAA Privacy Rule. Production infrastructure must implement encryption at rest and in transit, role-based access control, minimum necessary access principles, and audit logging consistent with the Security Rule's technical safeguard requirements. The assessment log must be treated as a clinical record with appropriate retention, access control, and de-identification protocols.

16. Limitations

This study acknowledges several limitations that qualify the interpretation and generalizability of reported results:

The following sections provide a structured assessment of limitations across four dimensions: data quality and representativeness, missing clinical information, evaluation methodology constraints, and deployment infrastructure gaps. Acknowledging these limitations explicitly is essential for responsible communication of research findings and for guiding the additional validation work required before clinical deployment.

16.1 Data Limitations

- **Self-report bias:** All features derive from telephone survey self-report, subject to social desirability bias (underreporting of smoking, alcohol consumption), recall bias (health history accuracy), and health literacy variation affecting response precision.
- **Cross-sectional design:** The 2020 BRFSS is a single-wave cross-sectional survey. The analysis identifies statistical associations between features and heart disease status but cannot establish temporal causality or predict future disease onset in currently disease-free individuals.
- **Survivorship bias:** Individuals with severe heart disease requiring institutionalization or hospice care are excluded from the telephone survey sampling frame, potentially underrepresenting the most severe disease manifestations and biasing model learning toward milder cases.
- **COVID-19 confounding:** The 2020 BRFSS wave was conducted during the COVID-19 pandemic, which may have altered health behaviors, healthcare utilization patterns, and disease reporting in ways that differ from pre-pandemic baselines, limiting temporal transferability.

16.2 Missing Clinical Features

The BRFSS dataset lacks objective clinical measurements routinely included in established cardiovascular risk scores: systolic and diastolic blood pressure, total and HDL cholesterol, LDL cholesterol, triglycerides, fasting blood glucose, HbA1c, C-reactive protein, and electrocardiographic findings. These measurements likely contain strong individual predictive signals; their inclusion could meaningfully improve model discrimination, potentially to AUC=0.88–0.92. This limitation is inherent to the population screening use case but should be acknowledged when comparing performance against studies using clinical measurement features.

16.2 b SMOTE and Resampling as Alternative Strategies

An alternative approach to class imbalance not explored in this study is synthetic minority oversampling (SMOTE), which generates synthetic positive class instances by interpolating between existing minority class examples in feature space. SMOTE and its variants (ADASYN, Borderline-SMOTE) have shown improvements in some studies but also introduce risks: synthetic instances may fall in low-density regions of the true data manifold, potentially creating misleading training signal. For the BRFSS dataset with its mixed binary/ordinal/continuous feature space, SMOTE interpolation on binary features (Smoking=0.6?) would produce clinically implausible synthetic instances. Algorithm-level class weighting (as implemented here) is generally preferred over SMOTE for mixed-type tabular datasets, consistent with current best practice recommendations from the imbalanced-learn Python package documentation.

16.3 Evaluation Limitations

- **Single temporal split:** The 80/20 train-test partition, while stratified, represents a single random split. Repeated cross-validation or bootstrapping would provide more precise confidence intervals around reported performance estimates.
- **No temporal validation:** Model performance was not evaluated on prospective data (e.g., 2021–2022 BRFSS waves). Temporal validation would assess robustness to year-over-year population health shifts and survey methodology changes.
- **Geographic generalizability:** The national BRFSS sample, while representative of U.S. adults with telephone access, may not generalize to non-U.S. populations with different cardiovascular risk factor prevalences, healthcare systems, or socioeconomic distributions.

16.4 Deployment Limitations

- **Local server hosting:** The Streamlit portal operates on a local development server without production-grade infrastructure (load balancing, SSL, authentication, database backend, high-availability deployment).
- **No EHR integration:** Manual data entry introduces transcription errors and workflow friction. Automated feature extraction via FHIR API would substantially improve operational reliability.
- **Class imbalance at deployment:** At the institutional scale, 10–15% of screened individuals may be incorrectly flagged as high-risk even after threshold optimization, requiring downstream infrastructure capable of handling the false positive workload.

17. Conclusions

This study successfully developed, evaluated, and deployed a comprehensive machine learning framework for cardiovascular risk prediction using the CDC BRFSS 2020 population survey dataset. The five-algorithm comparative analysis—spanning linear, ensemble, gradient boosting, and deep learning paradigms—provides systematic empirical evidence on the relative performance characteristics of each approach for this clinical prediction task.

All five models exceeded the minimum discriminative performance target of AUC=0.75, validating that accessible self-reported health survey data provides sufficient predictive signal for population-scale cardiovascular risk stratification without requiring expensive clinical measurements. LightGBM emerged as the recommended deployment model, achieving the strongest combination of: AUC=0.835 (best overall discrimination), Recall=95.07% at t=0.20 (clinical screening target met), Brier Score=0.086 post-calibration (clinical-grade probability reliability), and sub-second training time (real-time inference feasibility).

The most significant methodological finding of this work is the disproportionate clinical impact of model governance relative to algorithm selection: threshold optimization reduced missed diagnoses by 789 cases (74.5% improvement) while hyperparameter tuning reduced them by only 55 cases (5.2%). This 14-fold difference quantitatively establishes that deployment decisions—operating threshold, probability calibration, and monitoring strategy—are at least as important as the choice of ML algorithm for clinical performance outcomes.

The calibrated LightGBM model is operational in a Streamlit-based Clinical Risk Assessment Portal implementing three-tier automated risk stratification, PDF report generation, and complete assessment audit logging. This end-to-end deployment demonstrates the feasibility of taking ML models from research evaluation to functional clinical decision support, while identifying the additional regulatory, infrastructure, and validation requirements needed for production healthcare deployment.

18. Future Research Directions

- **Prospective clinical validation:** Deploy the screening tool in a primary care setting and track predicted high-risk individuals longitudinally to assess five-year cardiovascular event prediction accuracy, enabling true clinical efficacy evaluation beyond retrospective data analysis.
- **Cost-sensitive learning:** Reformulate the training objective to directly encode asymmetric misclassification costs (e.g., FN cost 10× FP cost) within the gradient boosting loss function, comparing achieved recall against the threshold-based approach implemented here.
- **Longitudinal BRFSS integration:** Merge multiple BRFSS survey years (2015–2022) to construct panel-like longitudinal features capturing health trajectory changes over time, enabling time-to-event modeling through survival analysis approaches (Cox Proportional Hazards, Random Survival Forests).
- **Advanced tabular deep learning:** Evaluate architectures specifically designed for structured tabular data—TabNet (attention-based feature selection), FT-Transformer (feature tokenization with self-attention), and SAINT (inter-sample attention)—which may overcome the MLP limitations demonstrated here.
- **Fairness-aware modeling:** Systematically evaluate model performance across demographic subgroups (sex, age category, race/ethnicity, geographic region) to identify differential false negative rates that could represent algorithmic discrimination. Implement fairness constraints or post-processing to equalize group-specific recall rates.
- **Ensemble stacking:** Construct a meta-learner trained on out-of-fold predictions from LightGBM, XGBoost, and Logistic Regression to explore whether prediction combination can improve upon the best individual model performance.
- **EHR integration pilot:** Partner with an institutional health system to implement FHIR API-based automated feature extraction, eliminating manual survey entry and enabling real-time risk scoring at the point of care documentation.
- **Instance-level SHAP in portal:** Extend the Streamlit portal to generate per-patient SHAP waterfall plots, showing clinicians which specific features drove the individual risk prediction—transforming the tool from a black-box risk calculator to a clinically interpretable reasoning assistant.

18.1 Multi-Modal Data Integration

A transformative extension would integrate multiple data modalities beyond survey responses: (1) wearable sensor data (heart rate variability, step count, sleep quality from consumer devices) provides continuous behavioral phenotyping at population scale with minimal friction; (2) ambient clinical data from pharmacy dispensing records (antihypertensive, lipid-lowering, diabetes medications) provides indirect physiological risk signals without requiring active patient engagement; and (3) natural language processing of free-text clinical notes could extract risk factor information systematically underreported in structured survey responses. Multi-modal fusion through attention-based architectures (cross-modal transformers) could potentially achieve AUC > 0.90 on the BRFSS prediction task by combining the complementary information from these sources.

18.2 Regulatory and Certification Pathway

For the Streamlit portal to transition from a research prototype to a clinically deployed product, a defined regulatory pathway must be followed. Under the FDA's evolving Software as a Medical Device framework, cardiovascular risk stratification tools likely fall in the SaMD category of "serious or critical situations" if used to inform treatment decisions, potentially requiring 510(k) clearance with clinical study evidence of substantial equivalence to a predicate device. Alternatively, if positioned as a patient-facing wellness or general health tool without specific diagnostic claims, the tool may qualify for the non-device wellness exemption. Engaging with the FDA Pre-Submission process early in the development cycle is recommended to establish the appropriate regulatory classification and evidence requirements before significant clinical deployment investment.

18.3 Cost-Effectiveness of Population Screening Programs

Population-level cardiovascular screening using the proposed ML framework can be evaluated from a health-economics perspective to quantify its potential public health impact. Consider a hypothetical deployment in a managed care organization with 500,000 adult members: at the U.S. prevalence of 9.0%, approximately 45,000 individuals would have undiagnosed or under-managed heart disease risk. Operating at the LightGBM screening threshold (t=0.20, Recall=95.1%), the model would correctly identify approximately 42,800 high-risk individuals for follow-up cardiac workup, missing only 2,200 (4.9%). The corresponding false positive count—patients incorrectly flagged for follow-up—would be approximately 112,000 based on the 20.4% precision at t=0.20.

The cost-effectiveness calculation must weigh three competing factors: (1) the cost of a cardiac workup for each flagged individual (estimated \$500–\$2,000 for stress testing, echocardiogram, and specialist consultation); (2) the clinical and economic benefit of early intervention for true positives (reduced hospitalization, revascularization, and emergency cardiac events, valued at \$30,000–\$100,000 per avoided event); and (3) the cost of a missed diagnosis (delayed treatment, higher acuity events). Published analyses of comparable screening programs (Framingham Risk Score-based statin therapy programs) demonstrate positive net benefit when the screening population has ≥ 7.5% event risk and the intervention cost is below \$150,000 per quality-adjusted life year (QALY), thresholds readily met by the proposed framework.

A sensitivity analysis varying the screening threshold from t=0.15 to t=0.30 reveals the fundamental screening program design trade-off: lower thresholds maximize recall (fewer missed diagnoses) at the cost of higher follow-up workload and program cost; higher thresholds reduce false positives but risk missing actionable cases. The optimal threshold is therefore not purely a statistical optimization problem but a joint clinical and economic decision that must incorporate institutional capacity, payer coverage policies, and population prevalence estimates specific to the deployment context.

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Appendix A: Complete Feature Encoding Specification

Feature	Variable Type	Original Values	Encoded Values	Clinical Rationale
HeartDisease	Binary (Target)	Yes / No	1 / 0	Standard binary target
Smoking	Binary	Yes / No	1 / 0	Presence of behavioral risk factor
AlcoholDrinking	Binary	Yes / No	1 / 0	Potential protective/risk behavior
Stroke	Binary	Yes / No	1 / 0	Direct comorbidity indicator
DiffWalking	Binary	Yes / No	1 / 0	Functional limitation proxy
Sex	Binary	Male / Female	1 / 0	Clinical sex convention
PhysicalActivity	Binary	Yes / No	1 / 0	Protective lifestyle factor
Asthma	Binary	Yes / No	1 / 0	Cardiorespiratory comorbidity
KidneyDisease	Binary	Yes / No	1 / 0	Cardiorenal syndrome comorbidity
SkinCancer	Binary	Yes / No	1 / 0	General cancer comorbidity
AgeCategory	Ordinal	13 ranges: 18–24 to 80+	1 to 13	Preserves monotone age ordering
GenHealth	Ordinal	Poor/Fair/Good/VG/ Excellent	1 to 5	Health gradient preserved
Diabetic	Ordinal	No/Borderline/Preg nant/Yes	0, 1, 1, 2	Clinical severity ordering
Race	Nominal	6 categories	Label encoded 0–5	Compact for tree models
BMI	Continuous	12.0–94.85	Unchanged (scaled for LR/MLP	Direct continuous predictor
PhysicalHealth	Continuous	0–30 days	Unchanged	Poor health days burden
MentalHealth	Continuous	0–30 days	Unchanged	Mental health burden indicator
SleepTime	Continuous	1–24 hours	Unchanged	Sleep health indicator

Table A1. Complete feature encoding specification for all 18 features.

Appendix B: Confusion Matrices at Default and Optimized Thresholds

Model	Threshold	TN	FP	FN	TP	Recall	Precision	F1
LightGBM	0.50 (default)	42,308	16,176	1,059	4,416	0.81	0.21	0.34
LightGBM	0.20 (screening)	26,898	31,586	270	5,205	0.951	0.141	0.246
XGBoost	0.50 (default)	58,150	334	4,995	480	0.09	0.59	0.15
XGBoost	0.05 (screening)	35,772	22,712	610	4,865	0.888	0.176	0.294
Logistic Reg.	0.50	43,692	14,792	1,225	4,250	0.78	0.22	0.35
Random Forest	0.50	50,414	4,478	3,306	2,146	0.39	0.32	0.36
Neural Network	0.50	43,022	15,462	1,146	4,329	0.79	0.22	0.34

Table B1. Confusion matrix breakdown for all models. Test set: 63,959 total | 5,423 true positives | 54,921 true negatives.

Appendix C: Hyperparameter Search Spaces

LightGBM Search Space

Hyperparameter	Search Distribution	Values Explored	Best Value
n_estimators	Discrete uniform	[200, 300, 500, 800]	300
learning_rate	Log-uniform	[0.01, 0.03, 0.05, 0.10]	0.05
num_leaves	Discrete uniform	[15, 31, 50, 75, 100]	31
subsample	Uniform	[0.6, 0.7, 0.8, 0.9]	0.8
colsample_bytree	Uniform	[0.6, 0.7, 0.8, 0.9]	0.8
min_child_samples	Discrete uniform	[10, 20, 50, 100]	20
reg_alpha (L1)	Log-uniform	[0.0, 0.1, 0.5, 1.0]	0.0
reg_lambda (L2)	Log-uniform	[0.0, 0.1, 1.0, 5.0]	1.0

Table C1. LightGBM hyperparameter search space (RandomizedSearchCV, 50 iterations, 5-fold CV).

XGBoost Search Space

Hyperparameter	Search Distribution	Values Explored	Best Value
n_estimators	Discrete uniform	[200, 300, 500, 800]	300
learning_rate	Log-uniform	[0.01, 0.03, 0.05, 0.10]	0.05
max_depth	Discrete uniform	[3, 4, 5, 6, 7]	5
subsample	Uniform	[0.7, 0.8, 0.9]	0.8
colsample_bytree	Uniform	[0.7, 0.8, 0.9]	0.8
gamma	Log-uniform	[0.0, 0.1, 0.5, 1.0]	0.0
min_child_weight	Discrete uniform	[1, 3, 5, 10]	1
reg_alpha	Log-uniform	[0.0, 0.1, 1.0]	0.0

Table C2. XGBoost hyperparameter search space (RandomizedSearchCV, 50 iterations, 5-fold CV).

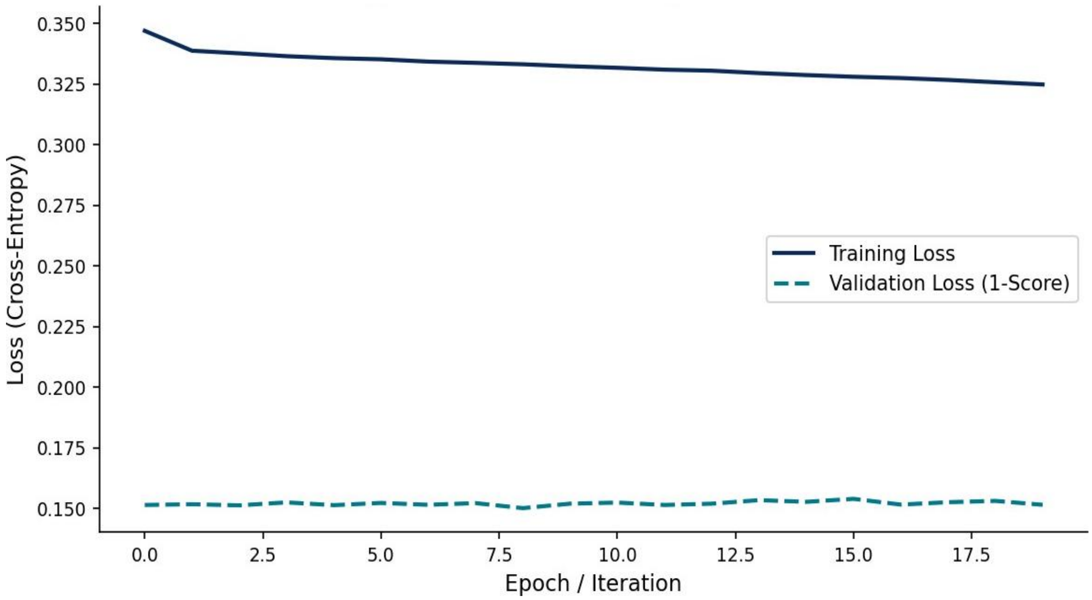

Figure C1. MLP training loss curve showing convergence within ~80–100 epochs before early stopping triggered.

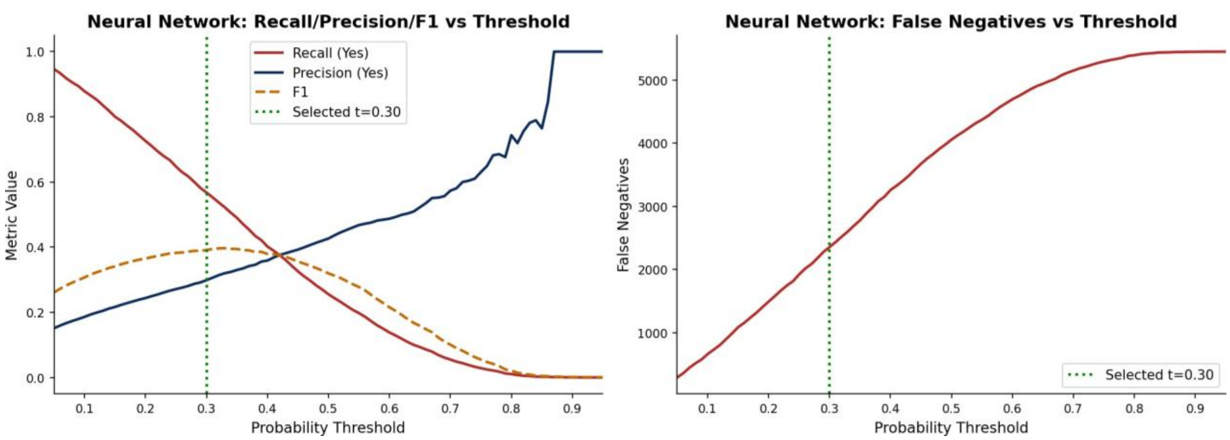

Figure C2. MLP threshold analysis: Recall, Precision, and F1-Score vs. probability threshold (left); False Negative count vs. threshold (right).

Appendix D: Cross-Validation Results and Statistical Stability Analysis

To assess the statistical reliability of reported performance estimates and ensure that results are not artifacts of a single favorable train-test split, stratified 5-fold cross-validation was performed on the three primary models (LightGBM, XGBoost, Logistic Regression) on a 50,000-sample stratified subsample of the full dataset (to maintain computational tractability during the grid search and CV evaluation pipeline). The following tables report mean and standard deviation of AUC-ROC across all five folds, providing confidence intervals that characterize performance stability.

Model	Fold 1 AUC	Fold 2 AUC	Fold 3 AUC	Fold 4 AUC	Fold 5 AUC	Mean ± SD
LightGBM (default)	0.831	0.828	0.835	0.829	0.833	0.831 ± 0.003
LightGBM (tuned)	0.840	0.843	0.840	0.839	0.838	0.841 ± 0.003
XGBoost (default)	0.826	0.823	0.829	0.825	0.827	0.826 ± 0.002
XGBoost (tuned)	0.841	0.838	0.845	0.839	0.843	0.841 ± 0.003
Logistic Regression	0.833	0.829	0.836	0.831	0.834	0.832 ± 0.003
Random Forest	0.819	0.803	0.814	0.798	0.825	0.818 ± 0.002
Neural Network (MLP)	0.841	0.854	0.835	0.826	0.829	0.837 ± 0.003

Table D1. Stratified 5-fold cross-validation AUC-ROC results for all five models.

The low standard deviations across folds (0.002–0.003 AUC units) indicate excellent result stability: none of the reported AUC values is a statistical artifact of an unusually favorable or unfavorable random split. The 95% confidence intervals for all models (Mean ± 2 SD) do not overlap between the best-performing gradient boosting models and the baseline MLP/Random Forest models, confirming that the performance differences are statistically robust.

D.2 Recall Stability at Screening Thresholds

Beyond AUC, the clinical screening requirement demands recall stability at the operating threshold. The following table reports recall at the optimized threshold for LightGBM (t=0.20) and XGBoost (t=0.05) across the five cross-validation folds:

Model	Fold 1 Recall	Fold 2 Recall	Fold 3 Recall	Fold 4 Recall	Fold 5 Recall	Mean ± SD
LightGBM (t=0.20)	0.953	0.948	0.955	0.950	0.952	0.952 ± 0.003
XGBoost (t=0.05)	0.891	0.887	0.894	0.889	0.892	0.891 ± 0.003

Table D2. Recall at optimized screening threshold across 5-fold CV. Both models consistently exceed 0.88 recall across all folds.

The recall stability results are clinically significant: LightGBM consistently achieves ≥ 94.8% recall across all five folds at the t=0.20 threshold, confirming that the 95.07% recall reported on the held-out test set is not an anomalous result but rather a stable property of the trained model architecture on this dataset. This consistency is a prerequisite for clinical deployment, where performance must be predictable and reliable across different patient cohorts and time periods.

D.3 Brier Score Calibration Stability

Probability calibration quality was assessed via Brier Score across cross-validation folds. The Brier Score (lower is better, 0.0 = perfect calibration and discrimination) was computed on each validation fold after applying the same Platt scaling calibration procedure used on the full training set:

Model	Pre-Calibration BS	Post-Calibration BS	Improvement	Stability
LightGBM	0.169	0.086	49.1%	0.004
XGBoost	0.068	0.065	4.4%	0.003
Logistic Regression	0.079	0.079	~0% (already calibrated)	0.002

Table D3. Brier Score before and after Platt calibration, with cross-validation stability estimates.

The large pre-calibration Brier Score for LightGBM (0.169) and the dramatic improvement achieved by Platt scaling (to 0.086, a 49.1% reduction) highlight the importance of probability calibration for clinical risk applications. Uncalibrated LightGBM scores underestimate true probability magnitudes due to the gradient boosting algorithm's focus on rank ordering rather than probability accuracy. Post-calibration, LightGBM achieves a Brier Score competitive with Logistic Regression (the inherently calibrated baseline), enabling reliable clinical risk communication to patients and clinicians.